MASTER'S DEGREE IN PHARMACEUTICAL BIOTECHNOLOGY

From Genome to Drug – Exploring Bioinformatics & Omics Sciences

Lesson VI - Proteomics Analysis

Proteomics – *In silico* nanobody mutagenesis

Estevão Barcelos, PostDoc UNIPG

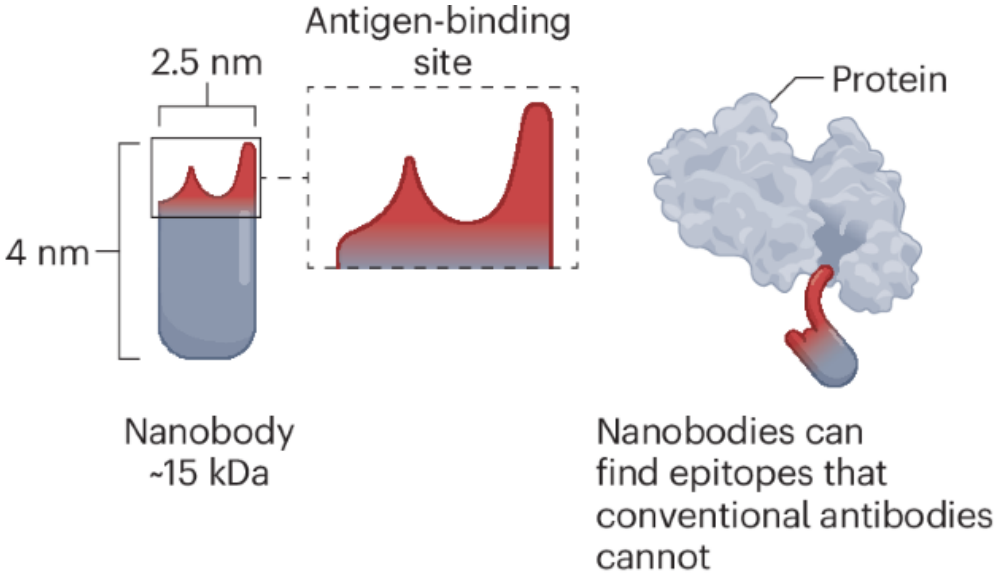
estevaocarlos.silvabarcelos@collaboratori.unipg.it

Learning Objectives — Lesson VI

Introduction T cell exhaustion and PD1

ΔΔG Interpretation

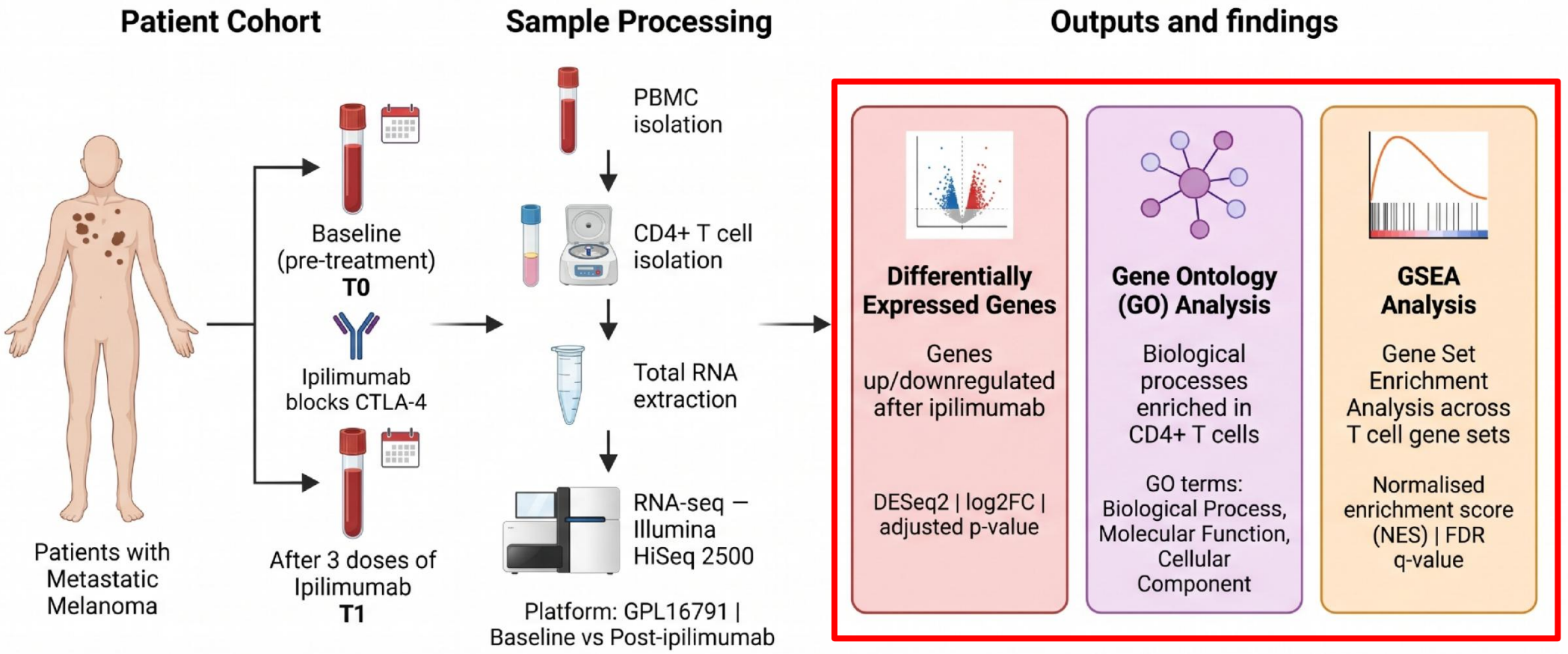
Nanobodies – AlphaFold3 prediction



Mutagenesis *in silico* with MutationExplorer

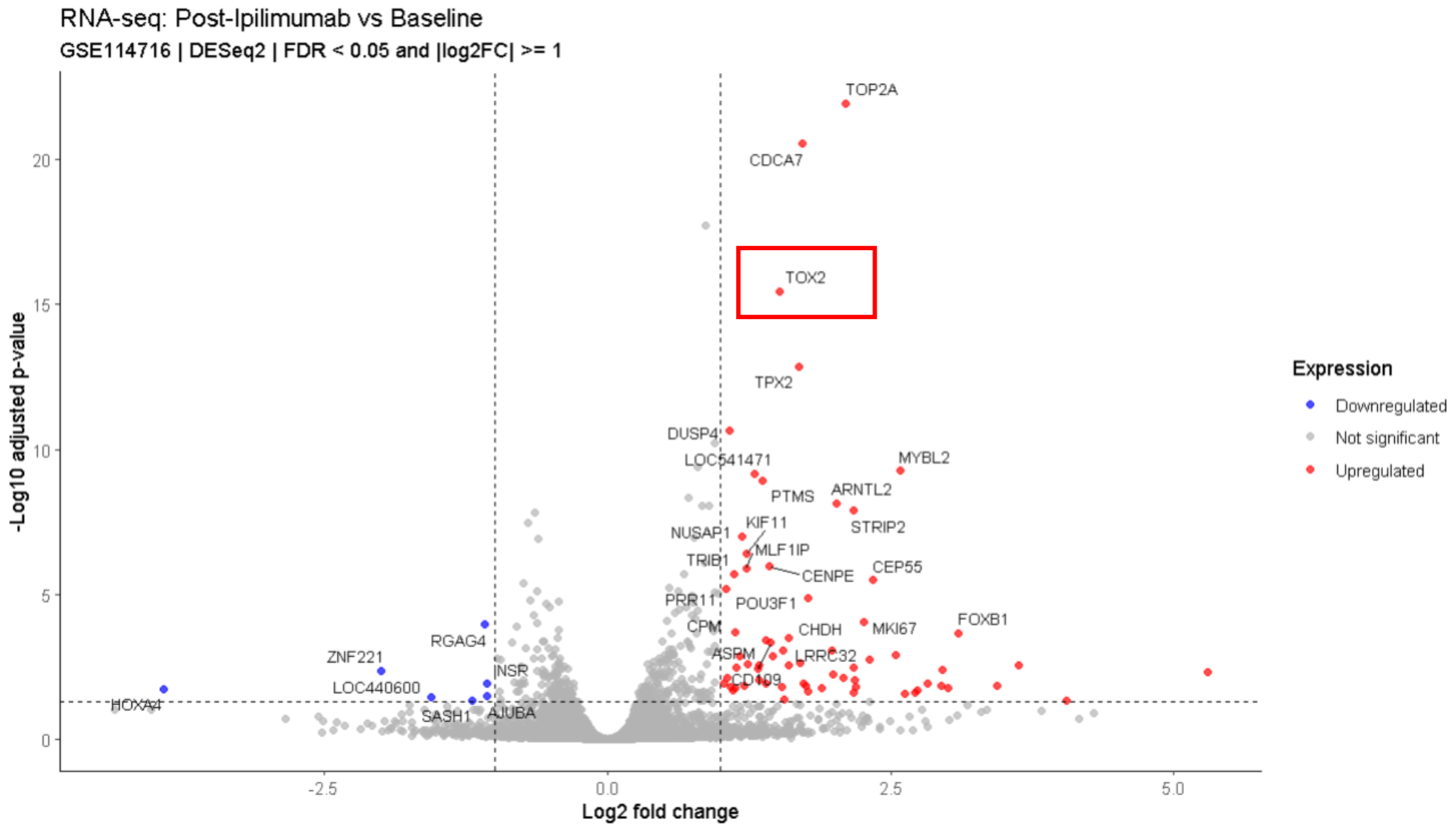
RNA-seq Analysis of CD4+ T cells following Ipilimumab Therapy

Goswami et al., 2018 — Journal of Clinical Investigation

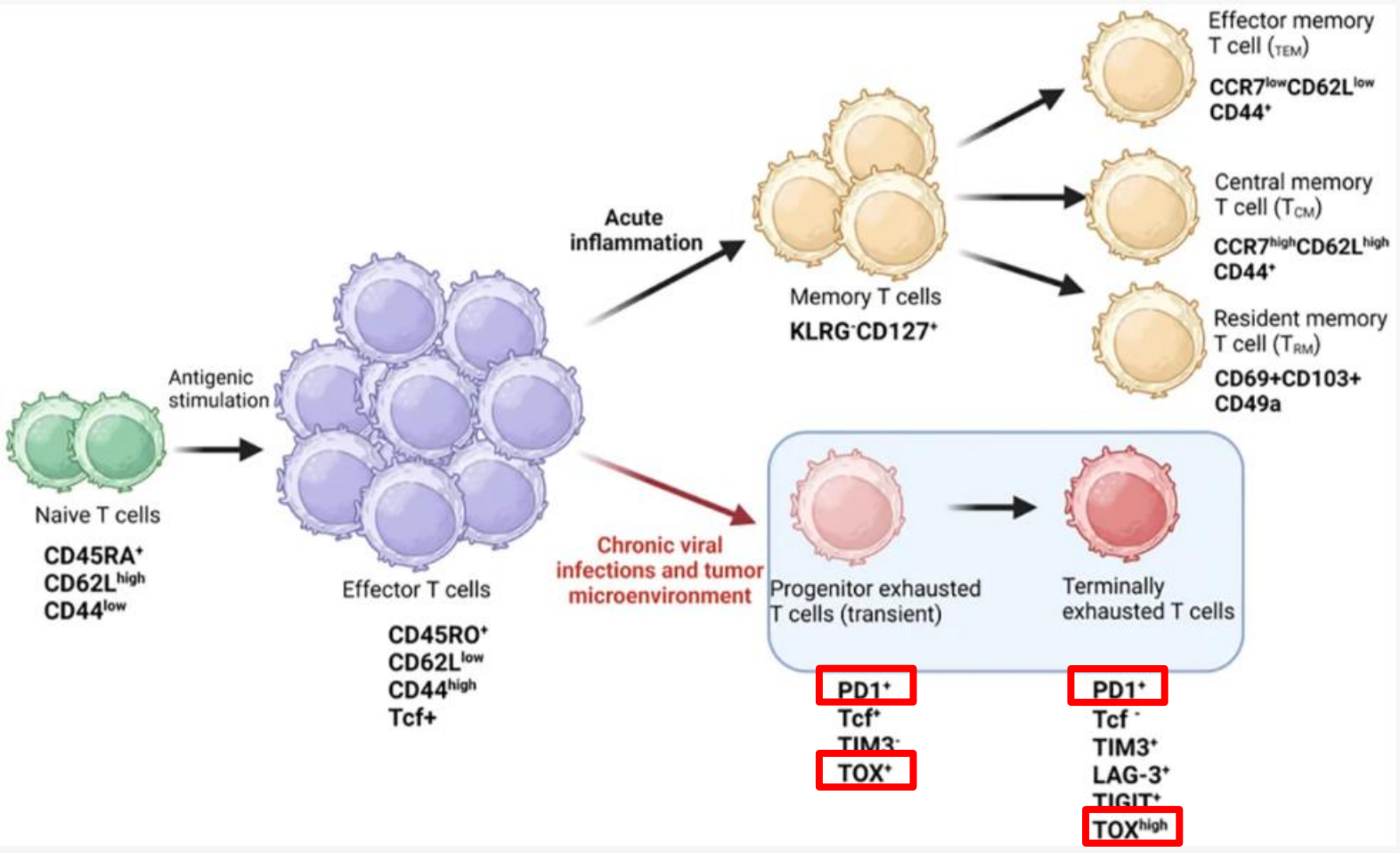


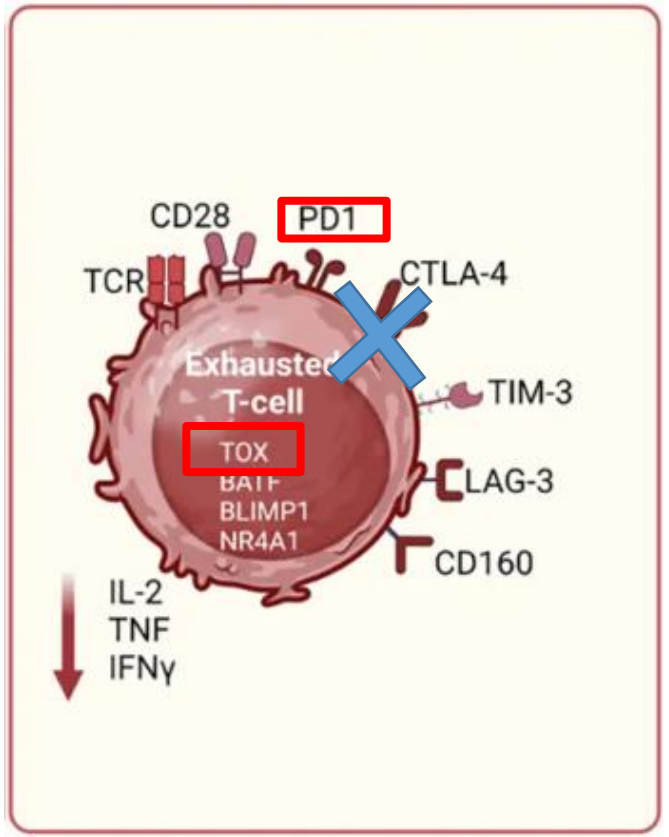
Goswami et al. (2018) JCI | GSE114716 | Illumina HiSeq 2500 | Homo sapiens — CD4+ T cells from PBMCs

High expression of *TOX2* after ipilimumab

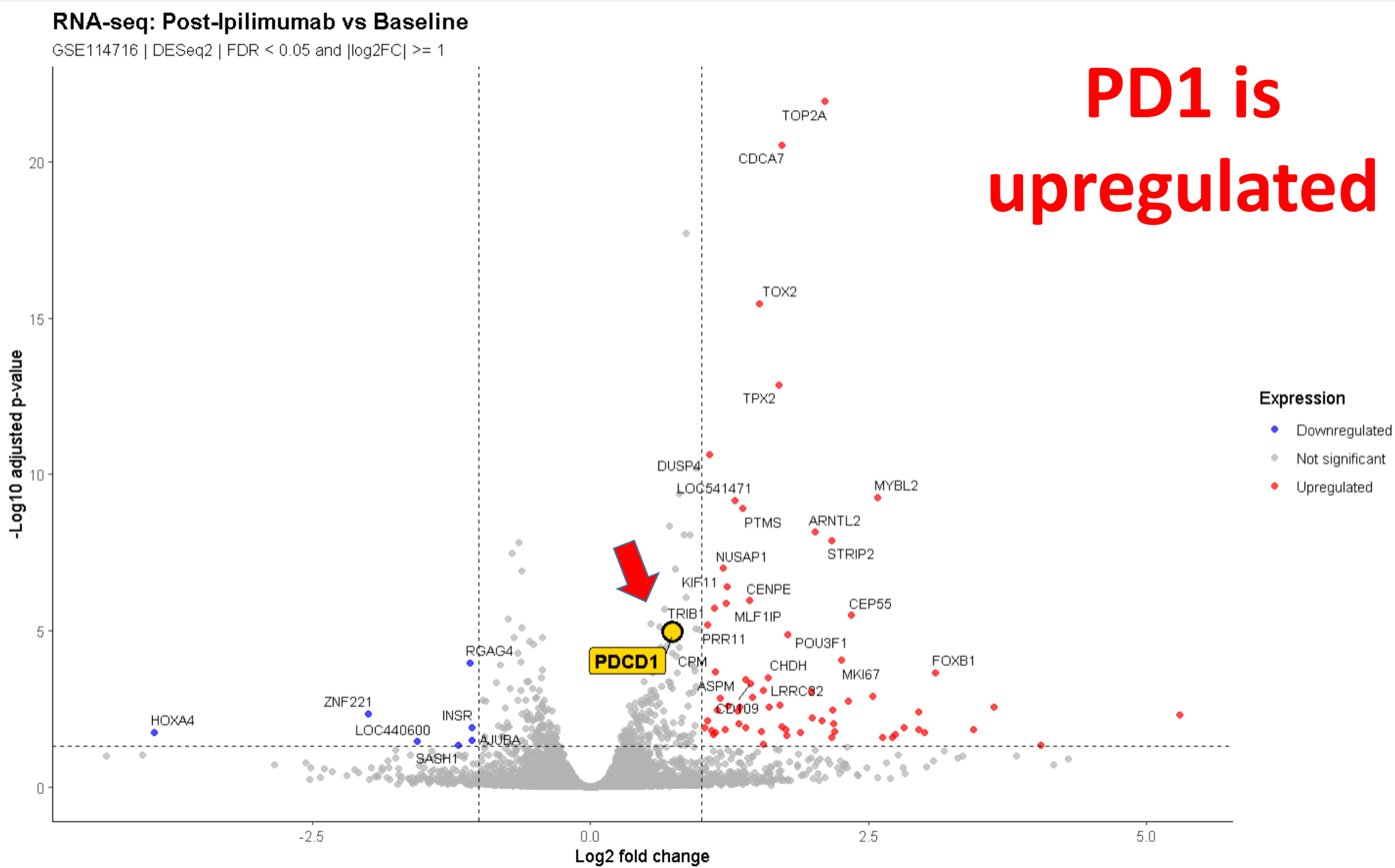


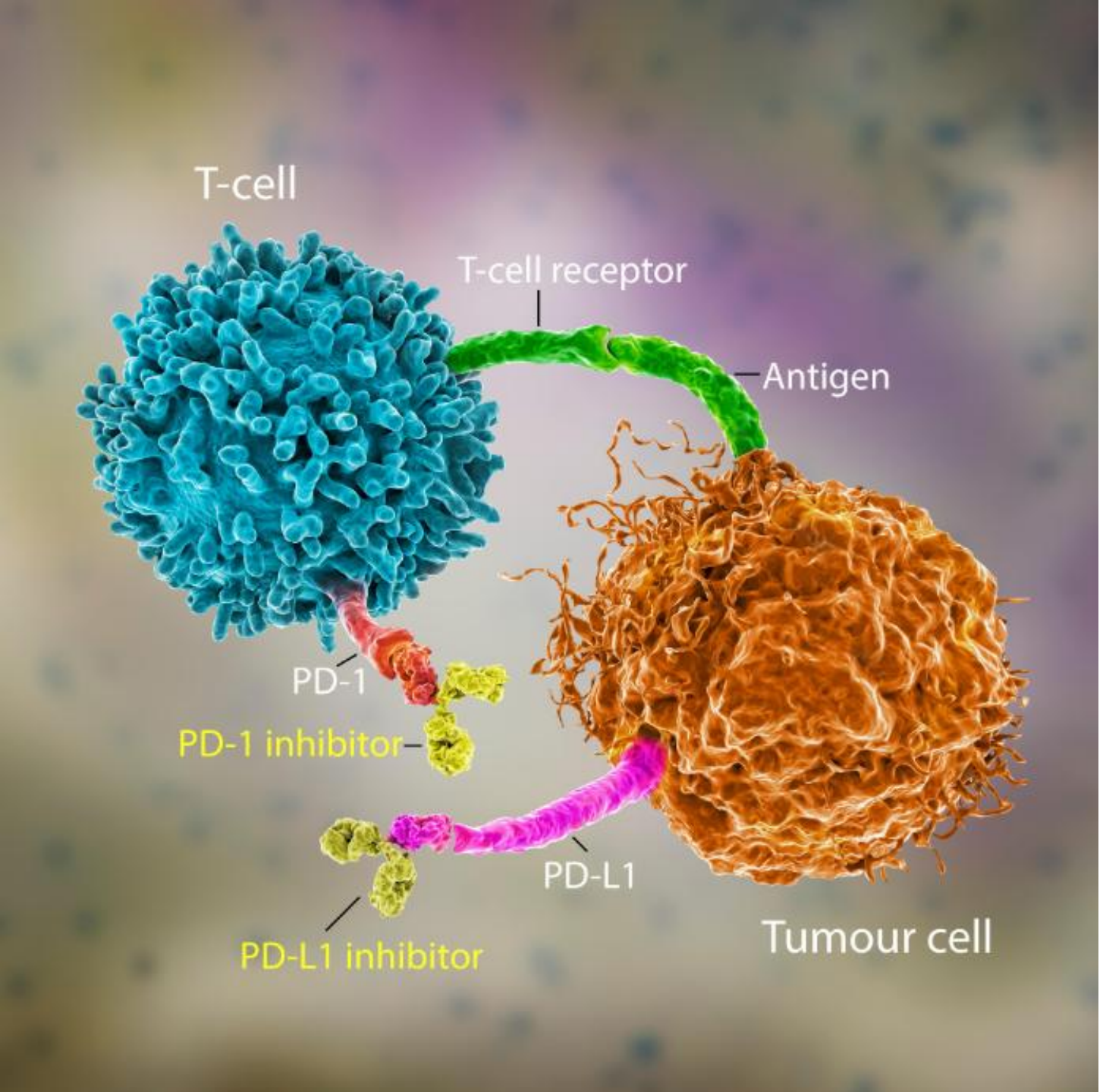
After activation





T cell Exhaustion

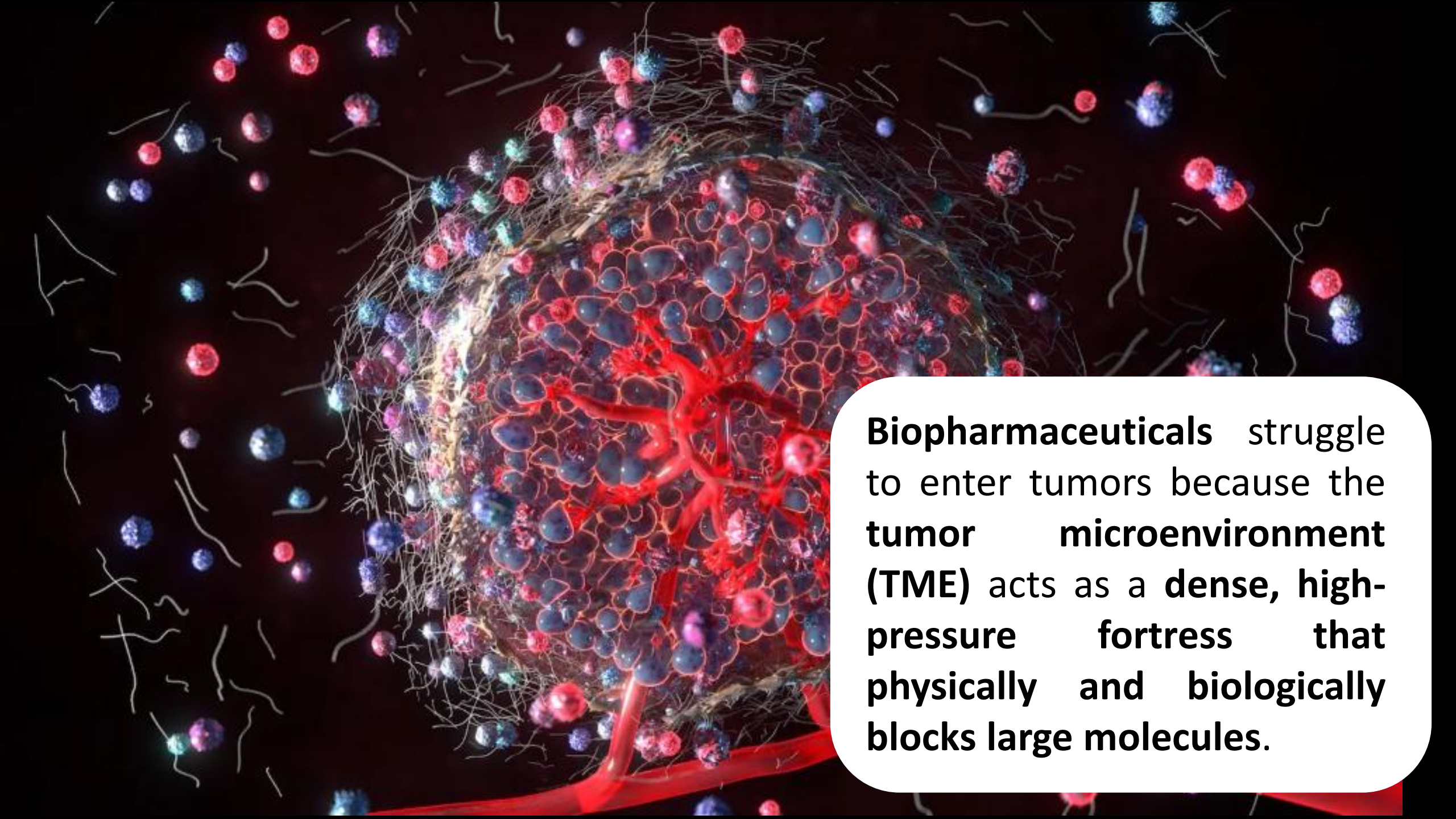

RNA-seq: Post-Ipilimumab vs Baseline
GSE114716 | DESeq2 | FDR < 0.05 and |log2FC| >= 1
PD1 is upregulated
-Log10 adjusted p-value
Log2 fold change
Expression
Downregulated
Not significant
Upregulated
PDCD1
TOP2A
CDCA7
TOX2
TPX2
DUSP4
LOC541471
MYBL2
PTMS
ARNTL2
STRIP2
NUSAP1
KIF11
CENPE
CEP55
TRIB1
MLF1IP
PRR11
POU3F1
RGAG4
CPM
CHDH
MKI67
FOXB1
ASPM
LRRC82
CD109
ZNF221
HOXA4
INSR
LOC440600
AJUBA
SASH1



PD-1 inhibitors are **monoclonal antibody (mAB)** drugs that block this interaction, reactivating exhausted T-cells so they can recognize and destroy cancer cells.

e.g. **Nivolumab**





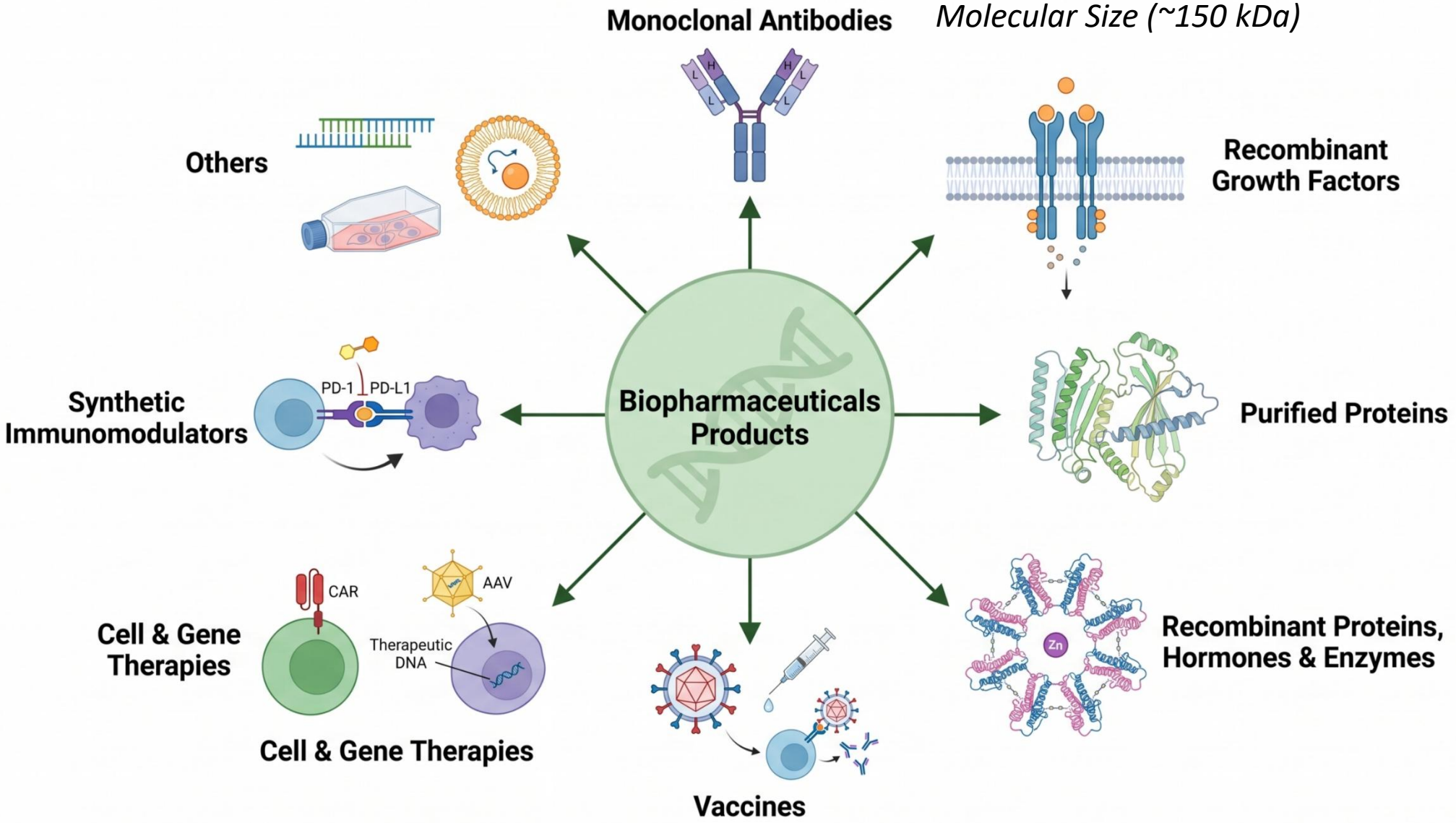
Biopharmaceuticals struggle to enter tumors because the **tumor microenvironment (TME)** acts as a **dense, high-pressure fortress that physically and biologically blocks large molecules**.



“We identified PD1 as a key driver of T cell exhaustion — but how do we drug a target hidden in tight cellular interfaces?

The answer may come from the immune system of camels — a single variable domain, ~15 kDa, capable of reaching where no antibody has gone before.

Which of these biopharmaceuticals can we use?



Penetrating the tumor microenvironment

Difficulty: high interstitial pressure, dense ECM, immunosuppressive milieu

✓ Best options for TME penetration

Vaccines
Small particles; systemic immune activation

Synthetic immunomodulators
Small molecules; checkpoint blockade (e.g. anti-PD-1/PD-L1)

Purified proteins
Smaller formats (nanobodies, cytokines) penetrate well

~ Moderate penetration

Recombinant proteins, hormones & enzymes
Moderate size; variable access

Recombinant growth factors
Can remodel TME vasculature (e.g. anti-VEGF), but limited entry

✗ Limited by size / require engineering

Monoclonal antibodies
~150 kDa — large; poor penetration unless bispecific or antibody fragments used

Cell & gene therapies
CAR-T cells struggle to traffic into solid tumors; major active research area

Others
Nanoparticles, exosomes being engineered for TME delivery

Key insight: size + immunological mechanism matter most

Small formats (vaccines, checkpoint inhibitors, cytokines) avoid the EPR & high IFP barriers.

Large biologics (mAbs, CAR-T) need engineering strategies: fragments, local delivery, or TME-remodeling.

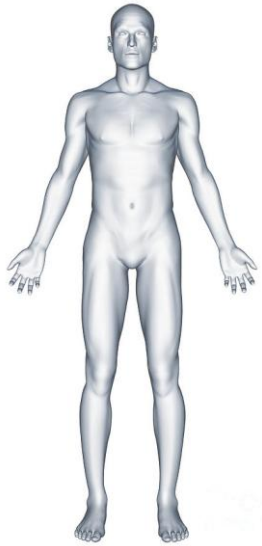
Combining approaches (e.g. anti-PD-L1 + vaccine) is now the dominant clinical strategy.

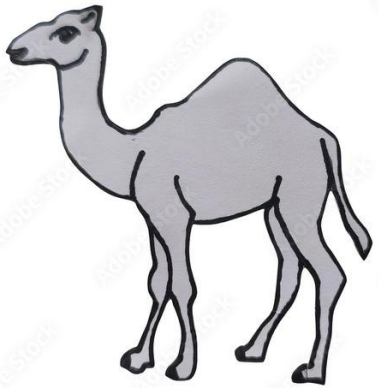
Best penetration Moderate Limited / needs engineering

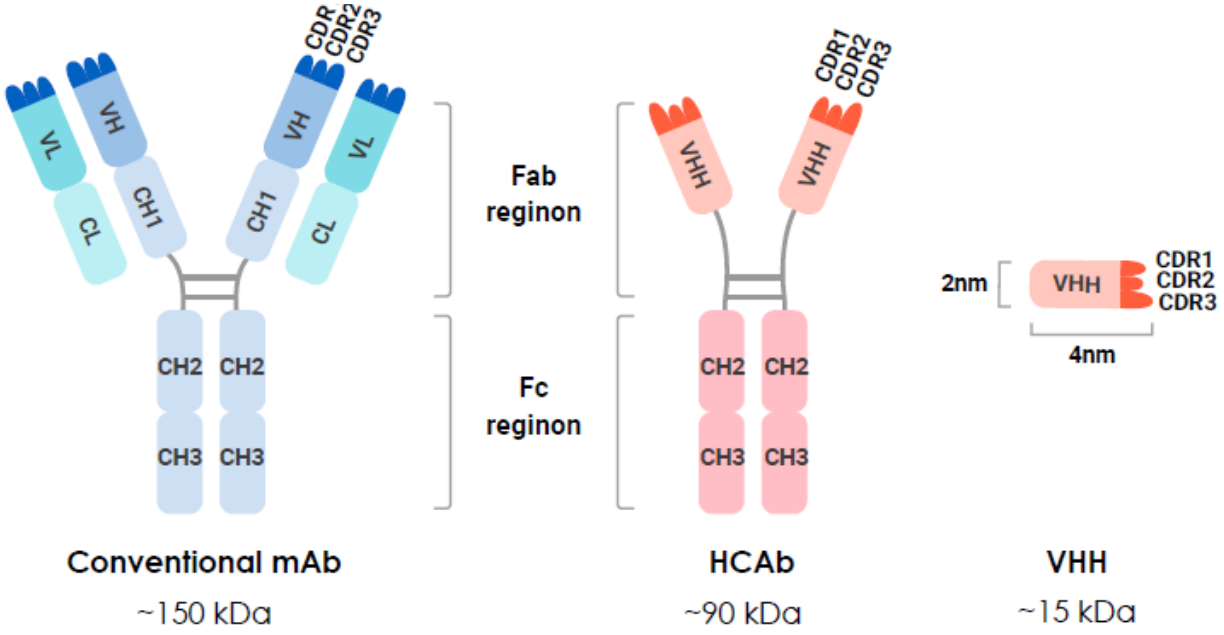
Another possibility…



Nanobodies







Conventional mAb

~150 kDa

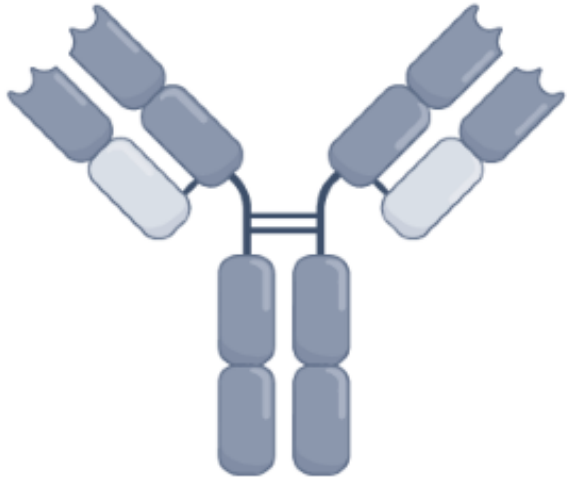
HCAb

~90 kDa

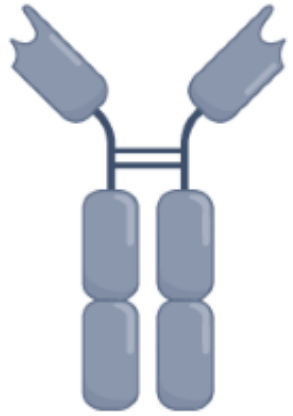
VHH

~15 kDa

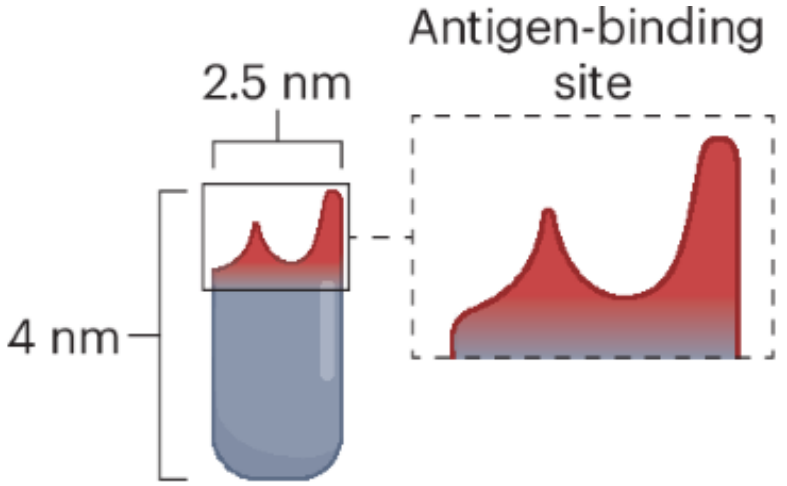
Why use Nanobodies?



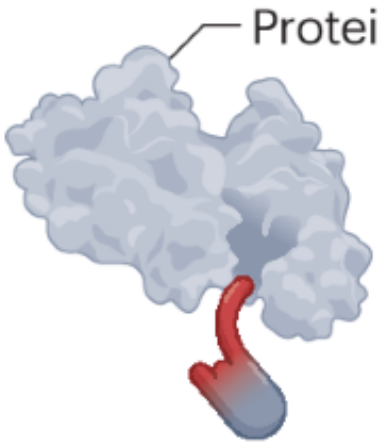
Immunoglobulin
~150 kDa



Heavy-chain-only
camelid antibody
~90 kDa



Nanobody
~15 kDa



Nanobodies can find epitopes that conventional antibodies cannot

- **Are small, <15 kDa**
- **Can be stored at room temperature**
- **Can reach epitopes where other nanobodies fail**
- **Are genetically simple and easy to produce**
- **Can be readily expressed and produced in *Escherichia coli***

What might the future hold?

- **Improved models of producing high-affinity binders**

Bioscience Reports (2023) **43** BSR20221546
https://doi.org/10.1042/BSR20221546

PORTLAND PRESS

Research Article

Screening and identification of an anti-PD-1 nanobody with antitumor activity

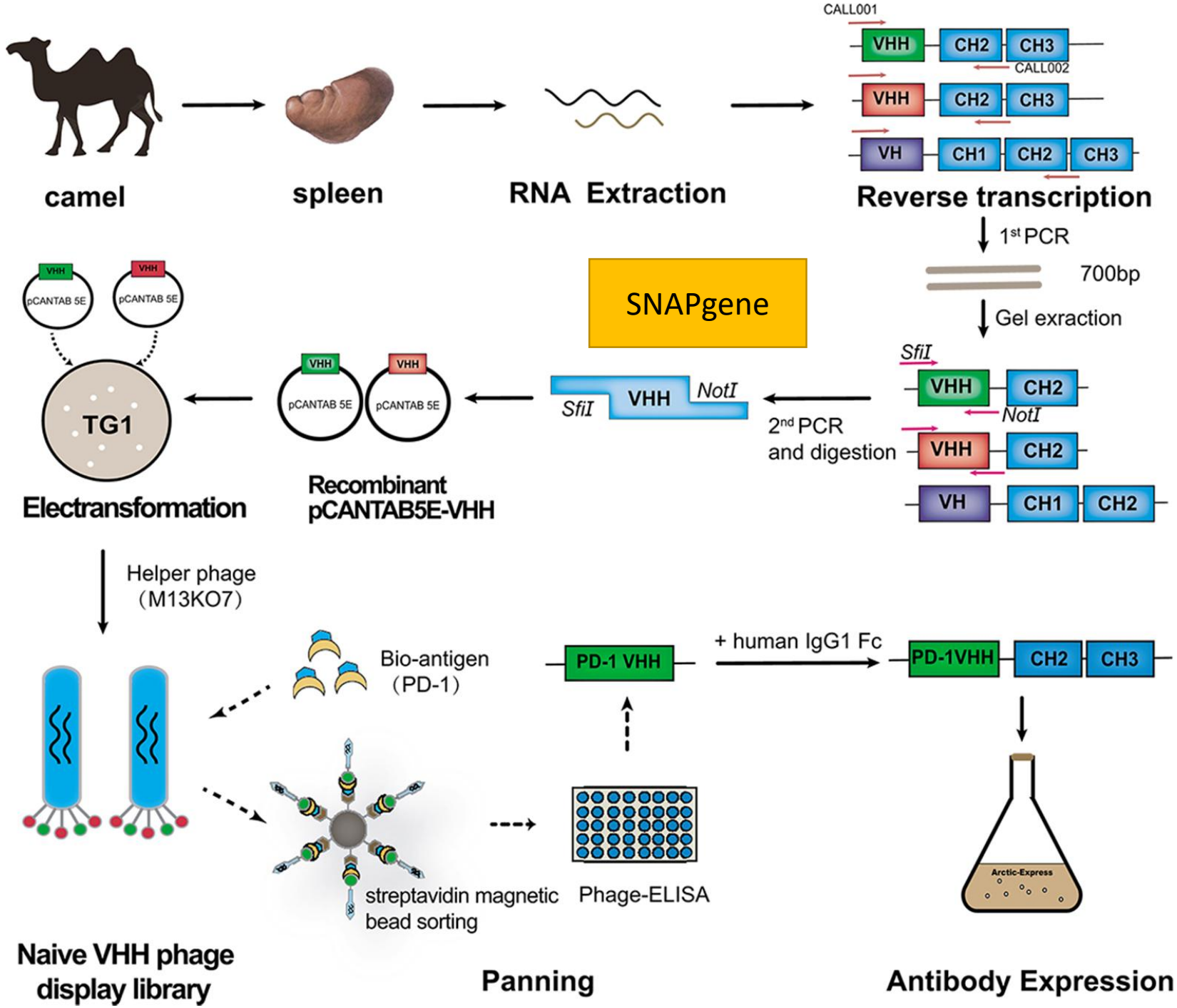
Yanting Zhang[1,2], Shaoqi Yang[1], Dan Jiang[3], Yanning Li[2], Shuo Ma[1], Liyan Wang[2], Guangqi Li[2], Hongxia Wang[2], Aijun Zhang[2] and Guangxian Xu[2,3]

[1]General Hospital of Ningxia Medical University, Yinchuan, Ningxia 750004, China; [2]Department of Laboratory Medicine, School of Clinical Medicine, Ningxia Medical University, Yinchuan, Ningxia 750004, China; [3]Guangdong Provincial Key Laboratory of Medical Molecular Diagnostics, The First Dongguan Affiliated Hospital, Guangdong Medical University, Dongguan, Guangdong 523808, China

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OPEN ACCESS

Blocking of programmed death protein 1 (PD-1) or its ligand PD-L1 with corresponding antibody to enhance T-cell response and mediate antitumor activity has been successfully applied in clinical practice. Several immune checkpoint inhibitors including monoclonal antibodies (mAbs) targeting PD-1 have been approved by the Food and Drug Administration (FDA) in cancer immunotherapy. However, the application of traditional antibodies has limited due to their drawbacks of large molecular weight (MW) and low tissue penetration. As the high specificity and strong tissue penetration of nanobodies (Nbs), efforts have been taken to develop Nbs for cancer therapy. Herein, we aim to screen a specific Nb against human PD-1 derived from a naïve camel Nb phage display library and further study its biological characteristic and antitumor activity. Finally, an anti-PD-1 Nb with high specificity and affinity was screened and generated, its cytotoxicity and antitumor effect was also confirmed *in vitro* and *in vivo*. All of these indicate that the anti-PD-1 Nb may provide an alternative and appealing therapeutic agent for cancer immunotherapy.

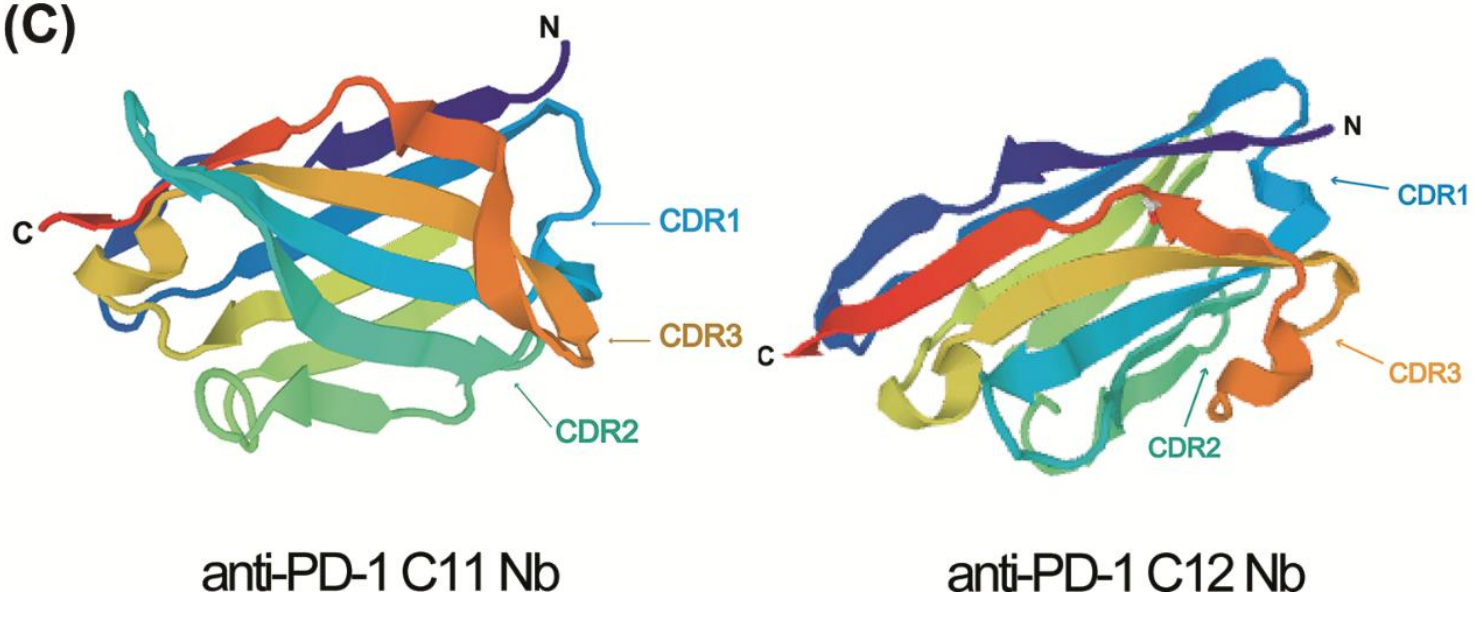

camel
spleen
RNA Extraction
CALL001
CALL002
VHH
CH2
CH3
VH
CH1
Reverse transcription
1st PCR
700bp
Gel exraction
SfiI
NotI
2nd PCR
and digestion
SNAPgene
pCANTAB 5E
TG1
Electransformation
Recombinant
pCANTAB5E-VHH
Helper phage
(M13KO7)
Bio-antigen
(PD-1)
PD-1 VHH
+ human IgG1 Fc
PD-1VHH
streptavidin magnetic
bead sorting
Phage-ELISA
Arctic-Express
Naive VHH phage
display library
Panning
Antibody Expression

Anti-PD-1 C11 and C12

(B)



(C)



(D)

Nanobody	MW（KD）	GRAVY
Anti-PD-1 C11 Nb	12.9	- 0.44
Anti-PD-1 C12 Nb	13.8	- 0.371

Pipeline

1. Clustal Omega to compare the sequences of Nanobodies

2. Retrieve the Extracellular sequence of PD1 from **UniProt**

3. AlphaFold3 prediction interaction between the **nanobody and PD-1**

4. Upload the .cif file into **ChimeraX**

5. Upload the .pdb file into **MutationExplorer**

6. Check the ΔΔG between the original sequence and the mutated sequence

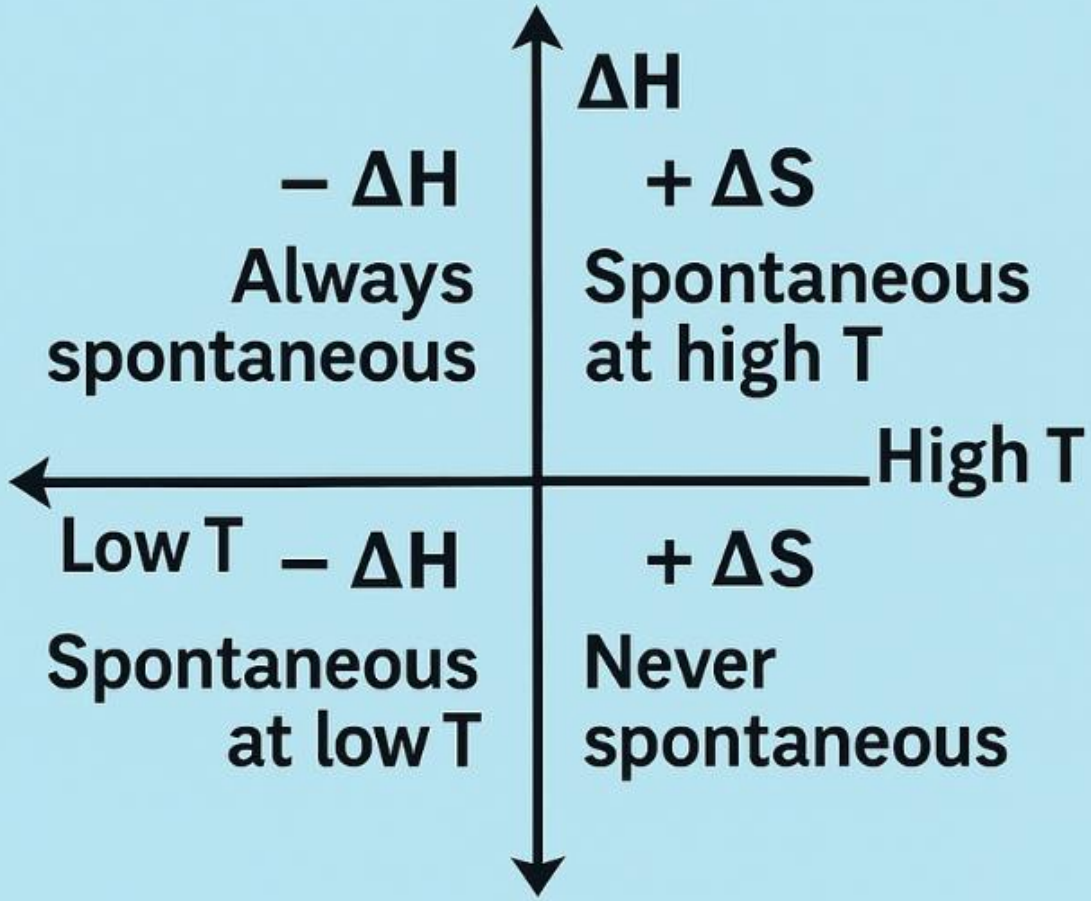
Delta delta G (ΔΔG or ddG)

Delta delta G (ΔΔG or ddG) is a thermodynamic metric that measures how an amino acid mutation changes an antibody's **binding affinity** or **structural stability**.

In antibody production and engineering, scientists use ΔΔG to predict which **genetic mutations will create stronger**, **more stable therapeutic antibodies before making them in the lab**.

GIBBS FREE ENERGY

ΔG, SPONTANEITY, AND EQUILIBRIUM



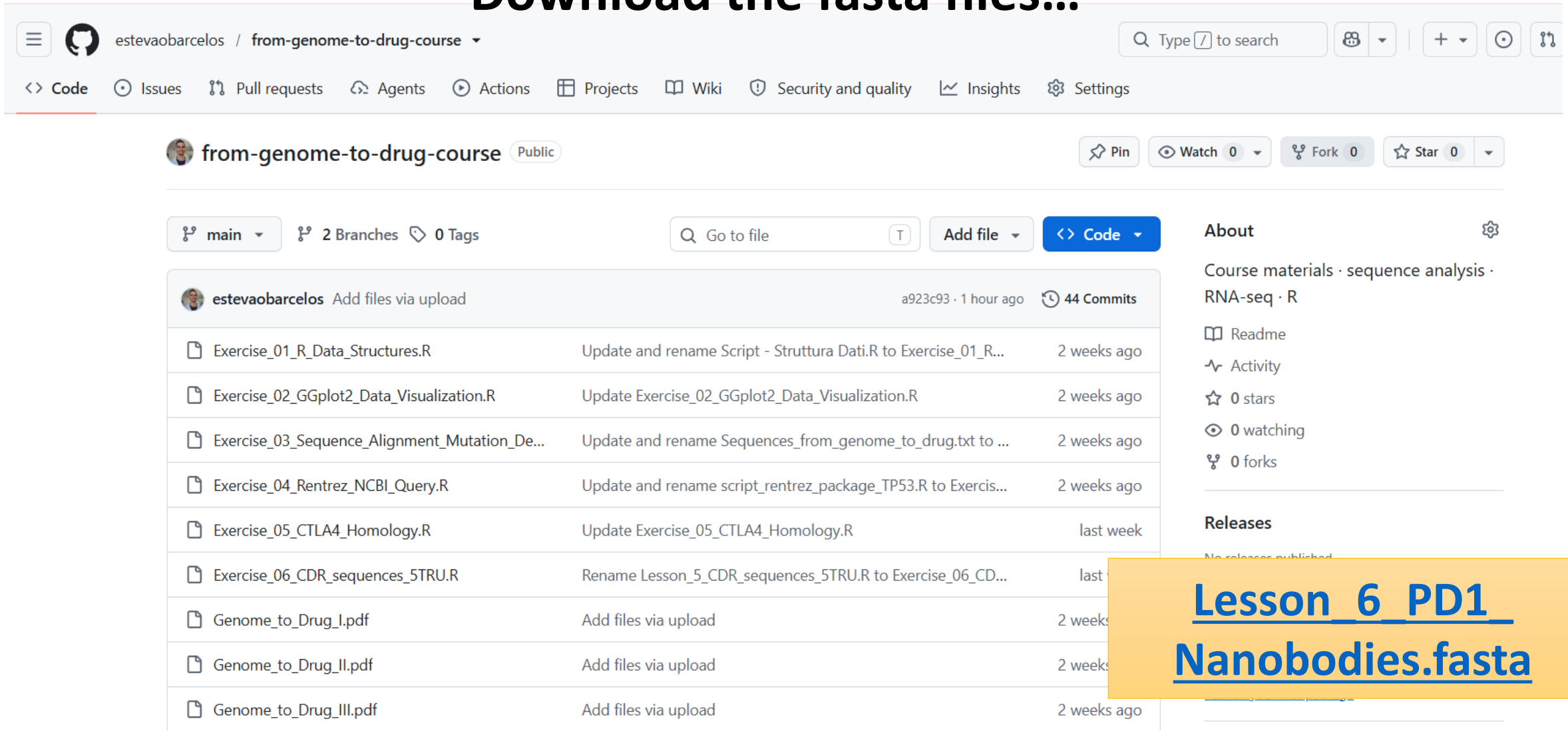
$\Delta G < 0$
Spontaneous

$\Delta G > 0$
Nonspontaneous

$\Delta G = 0$
At equilibrrium

https://github.com/estevaobarcelos/from-genome-to-drug-course

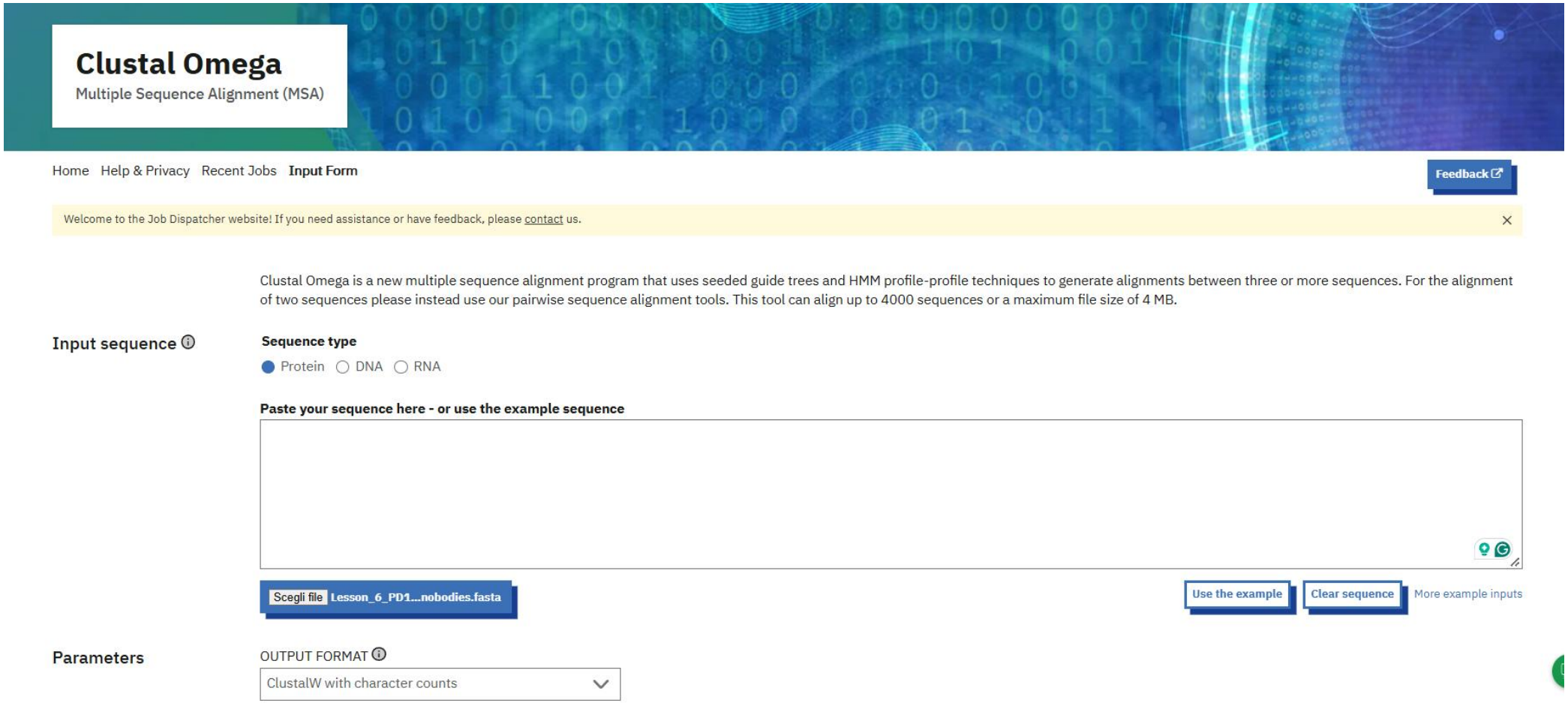
Download the fasta files…



Lesson_6_PD1_Nanobodies.fasta

Clustal Omega

Compare the sequences…



Results of comparison

Tool output

Download

```
CLUSTAL O(1.2.4) multiple sequence alignment


anti-PD-1_C11_Nb      QVQLQESGGAWCEPGGSLTLPCAVSGFAVDRDAVSWVRKAPGQGLQWVSTTPDGT--NYY      58
anti-PD-1_C12_Nb      QVQLQELGETRCRHGGSLRLSCKVSGYTQRSYCLGWLRQAPGNEREAVAVLQHSGNFRHY      60
                      ****** * : *. **** * * ***::    .:.*:*:***:  : *:.  ..   .:*

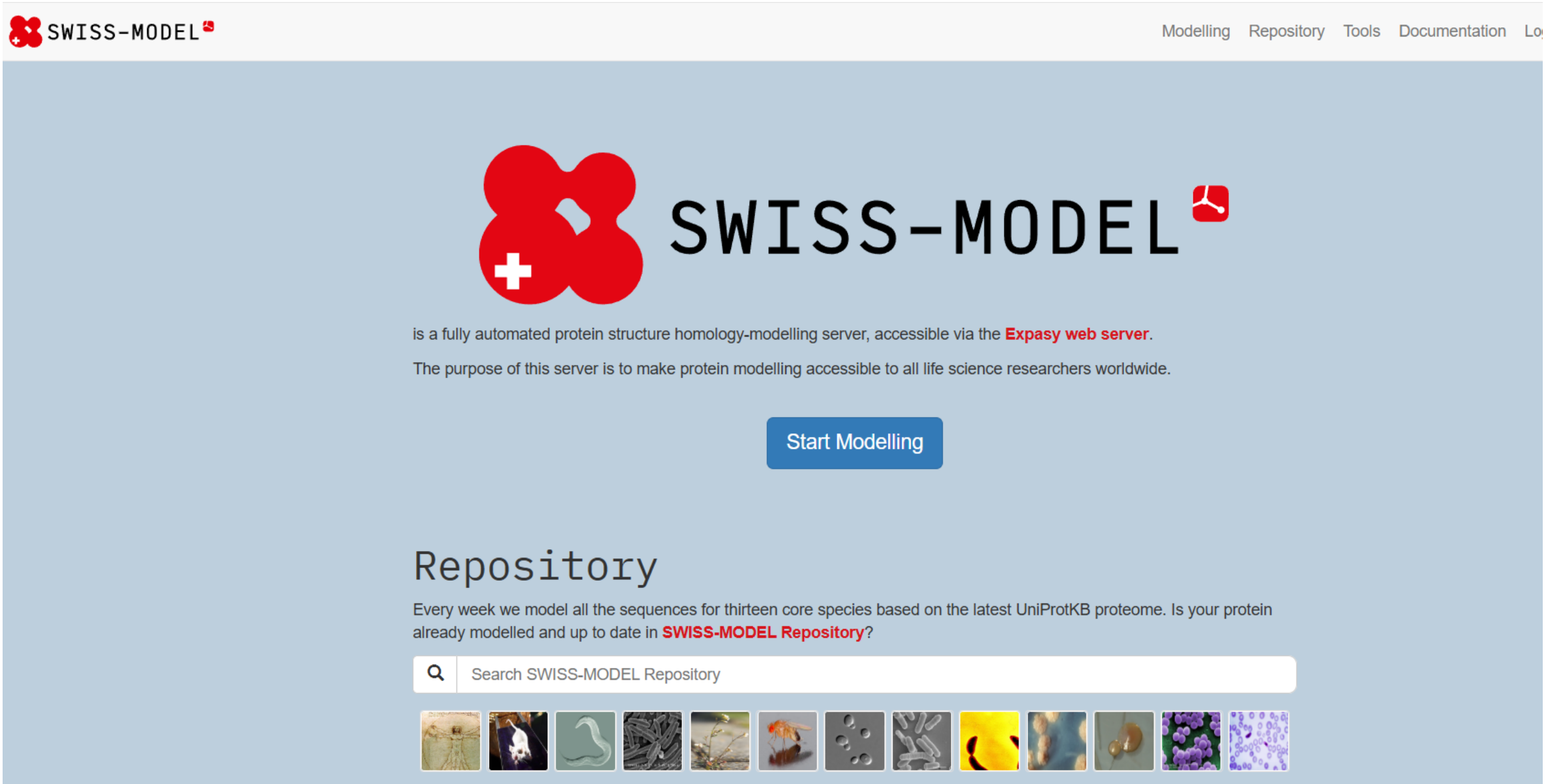

anti-PD-1_C11_Nb      GALAKDRFTVTRDTTRNTLQLQLTKLRIEDTAKYHCVQDYCSTGRCSLRTQGTQVTVSS      117
anti-PD-1_C12_Nb      ADSVQGRFVISQDYTDNTVYLAMNNLKSEDAAMYYCAAVGLRVSQFNYWGQGTQVTVSS      119
                      .  .:.**.:::* * **: * :.:*: **:* *:*.     ..: .   *********
```

Coloured sequences

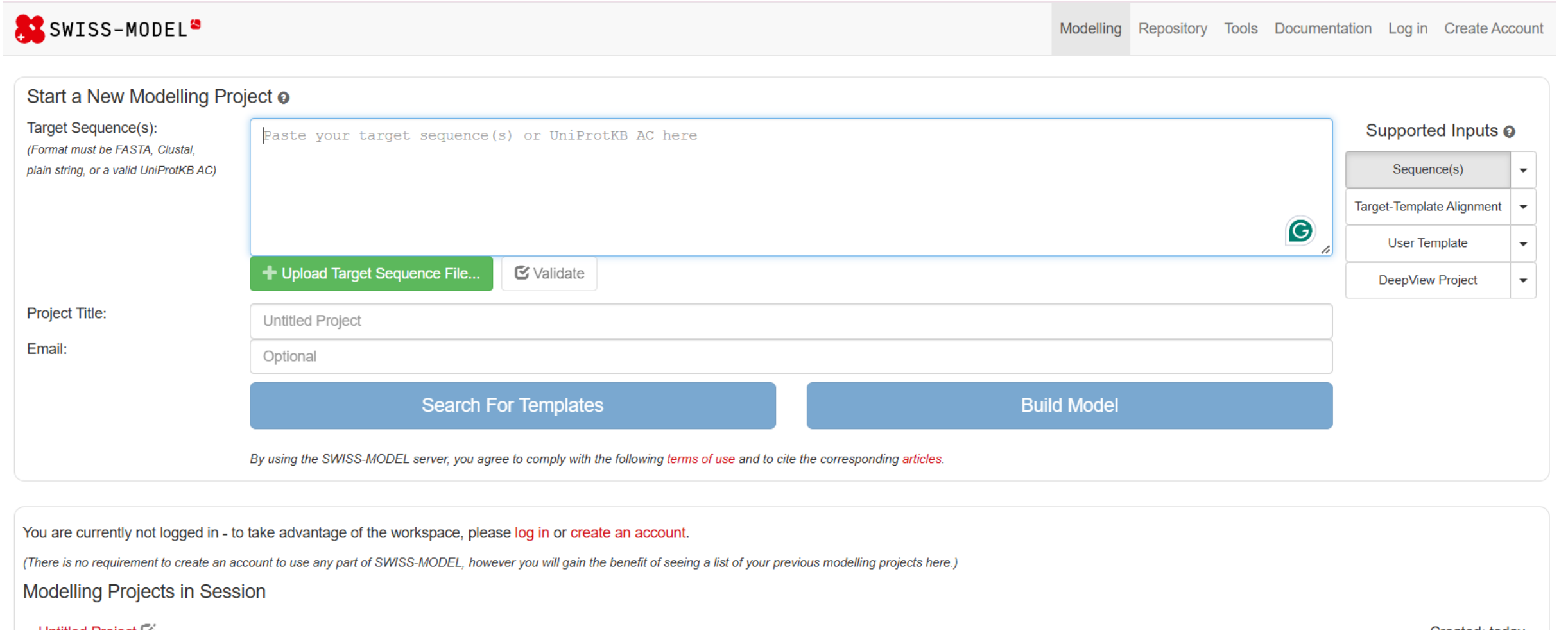
Show

*Identity
: highly similar properties
. "weakly similar" properties
unpreserved residues

https://swissmodel.expasy.org/



https://swissmodel.expasy.org/interactive



Swiss Model

SWISS-MODEL

Modelling | Repository | Tools | Documentation | Log in | C

All Projects

Untitled Project Created: today at 14:30

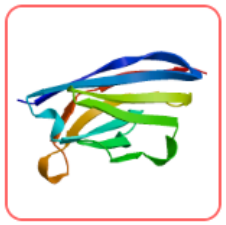
Summary | Templates 50 | Models 1 | Project Data

Model Results

Order by: GMQE

1 — 119

Model 01



Structure Assessment

Compare ☐

Download files | Display files

Oligo-State
Monomer

GMQE
0.78

QMEANDisCo Global:
0.78 ± 0.08

QMEANDisCo Local

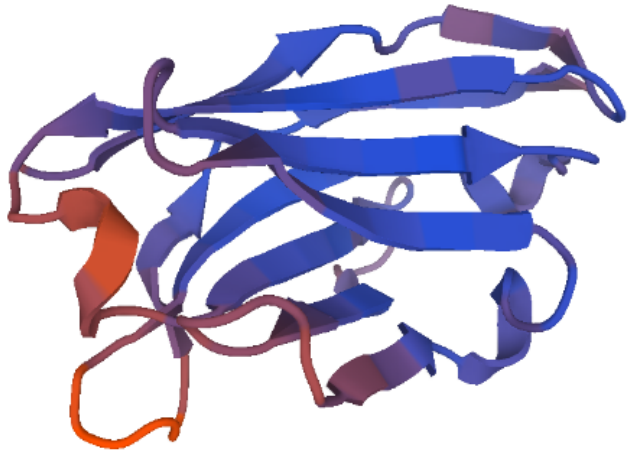
QMEAN Z-Scores

Template

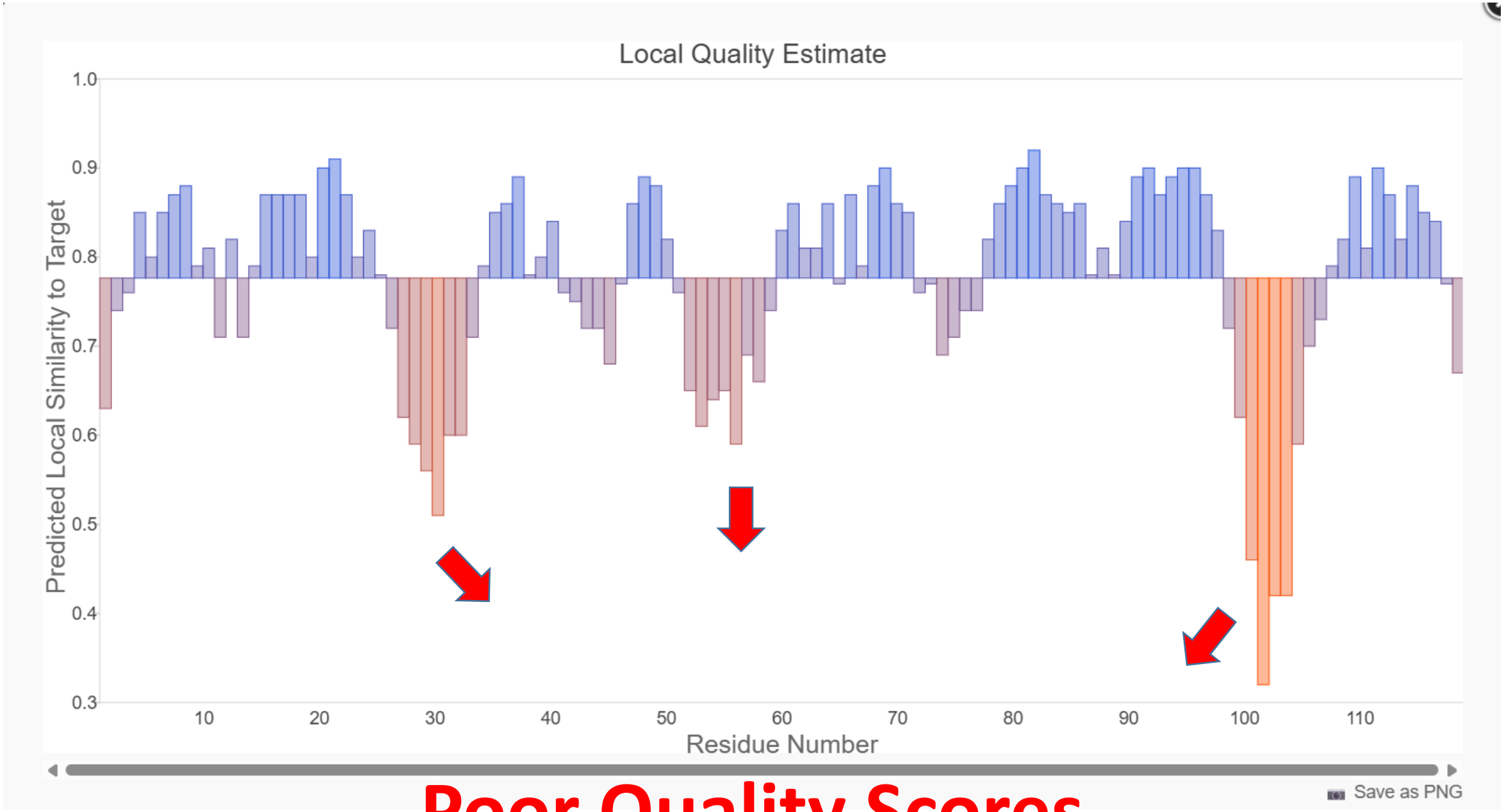
6waq.2.B nanobody SARS VHH-72

Crystal structure of the SARS-CoV-1 RBD bound by the cross-reactive single-domain antibody SARS VHH-72

Seq Identity
63.03%
Coverage



Confidence (gradient) | Carto



Poor Quality Scores

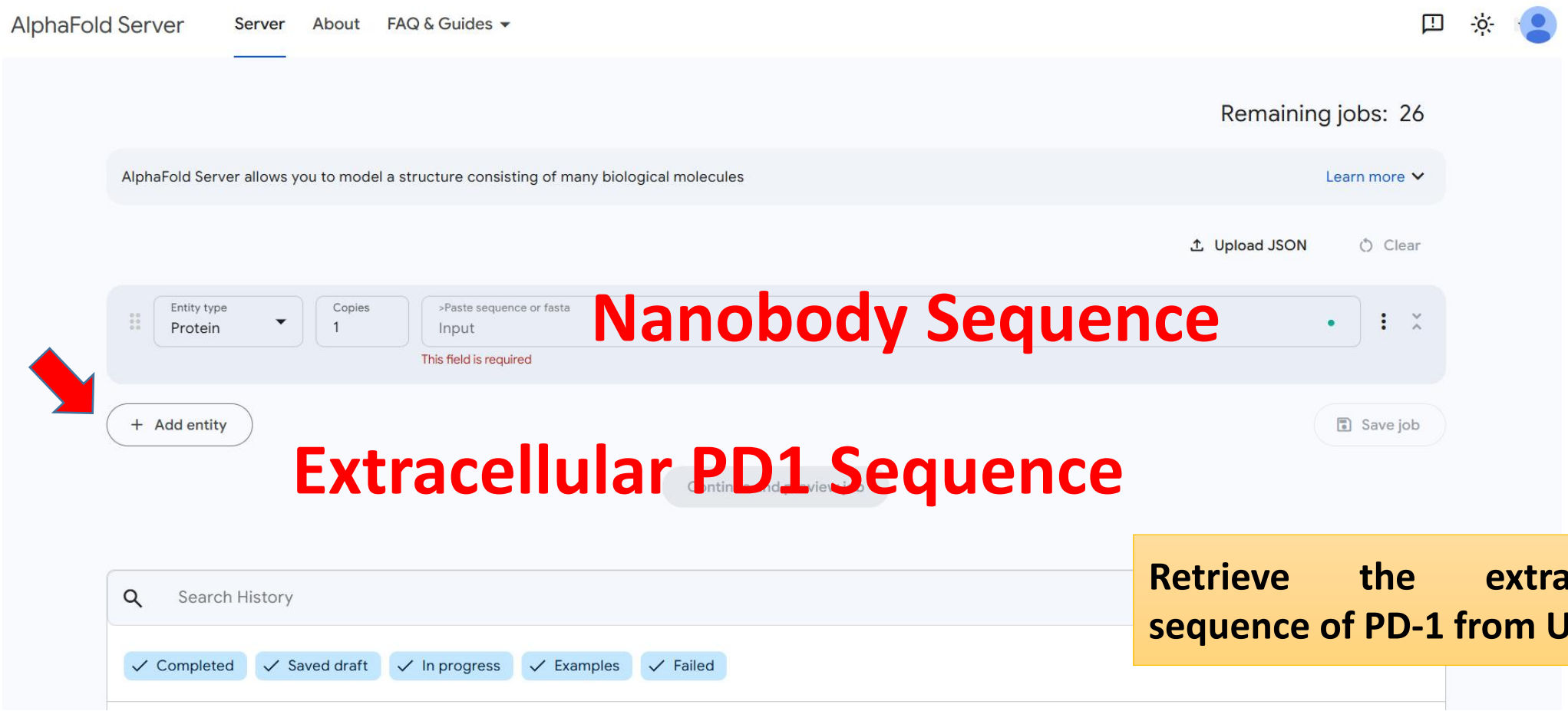
https://alphafoldserver.com/

Model the structure/interaction…



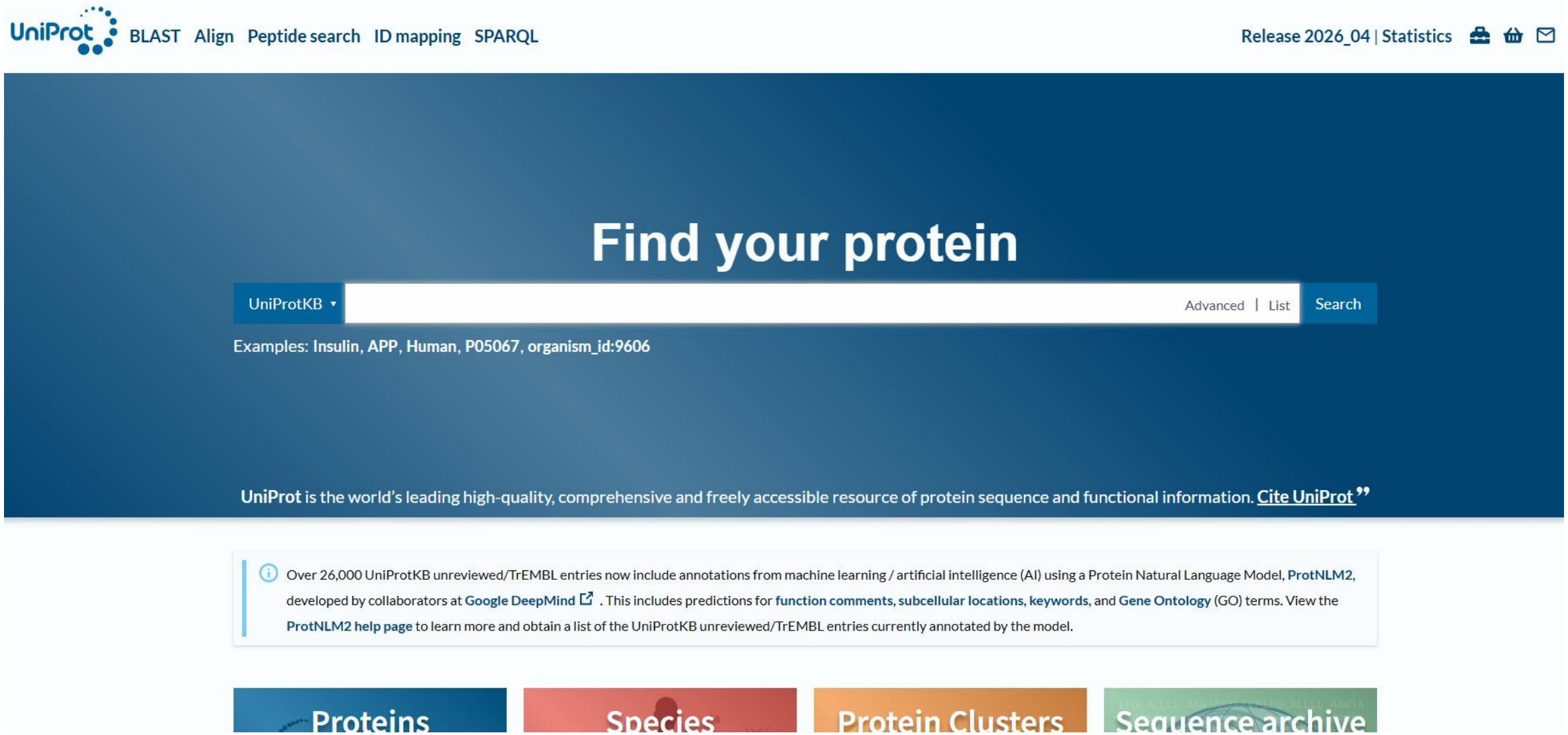
https://alphafoldserver.com/

Model the structure/interaction…

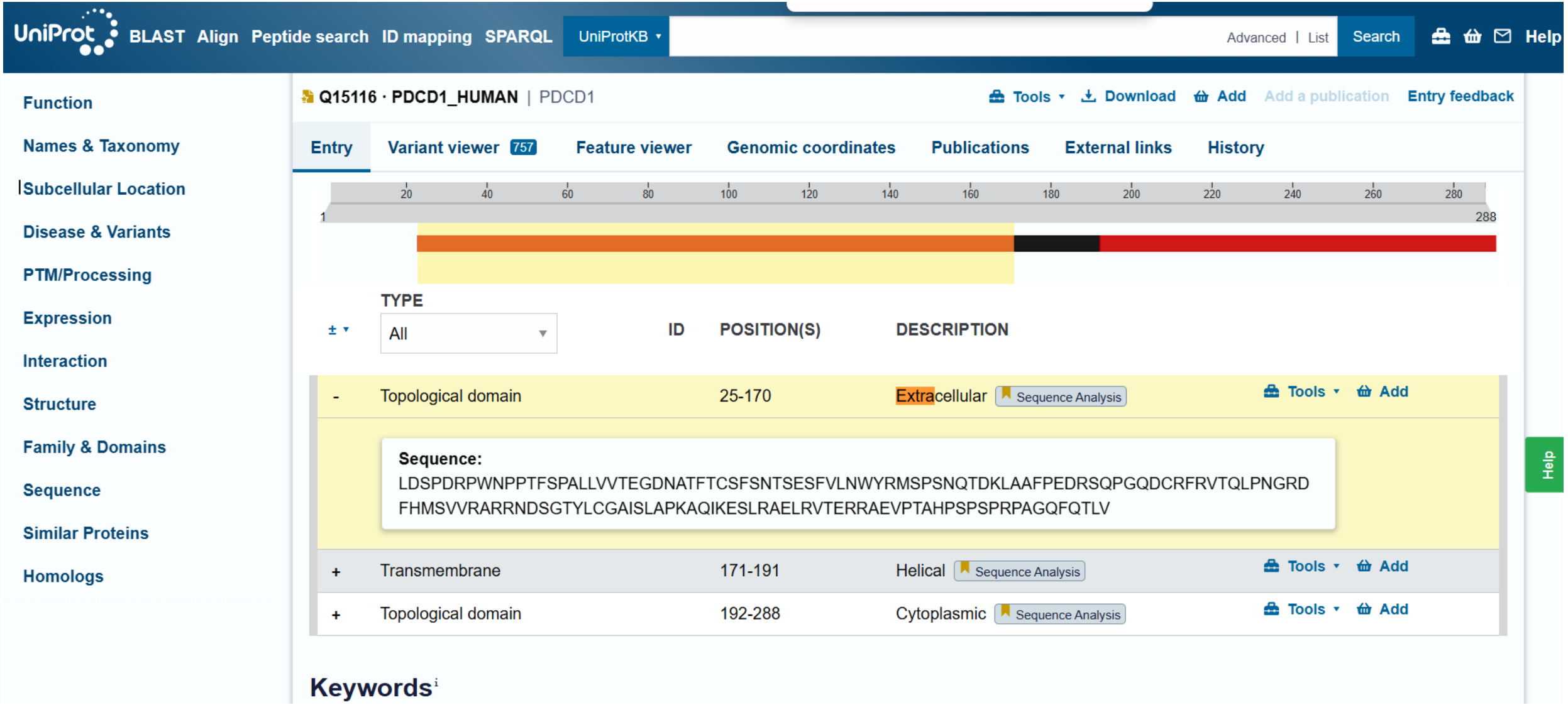


Retrieve the extracellular sequence of PD-1 from UniProt

https://www.uniprot.org/

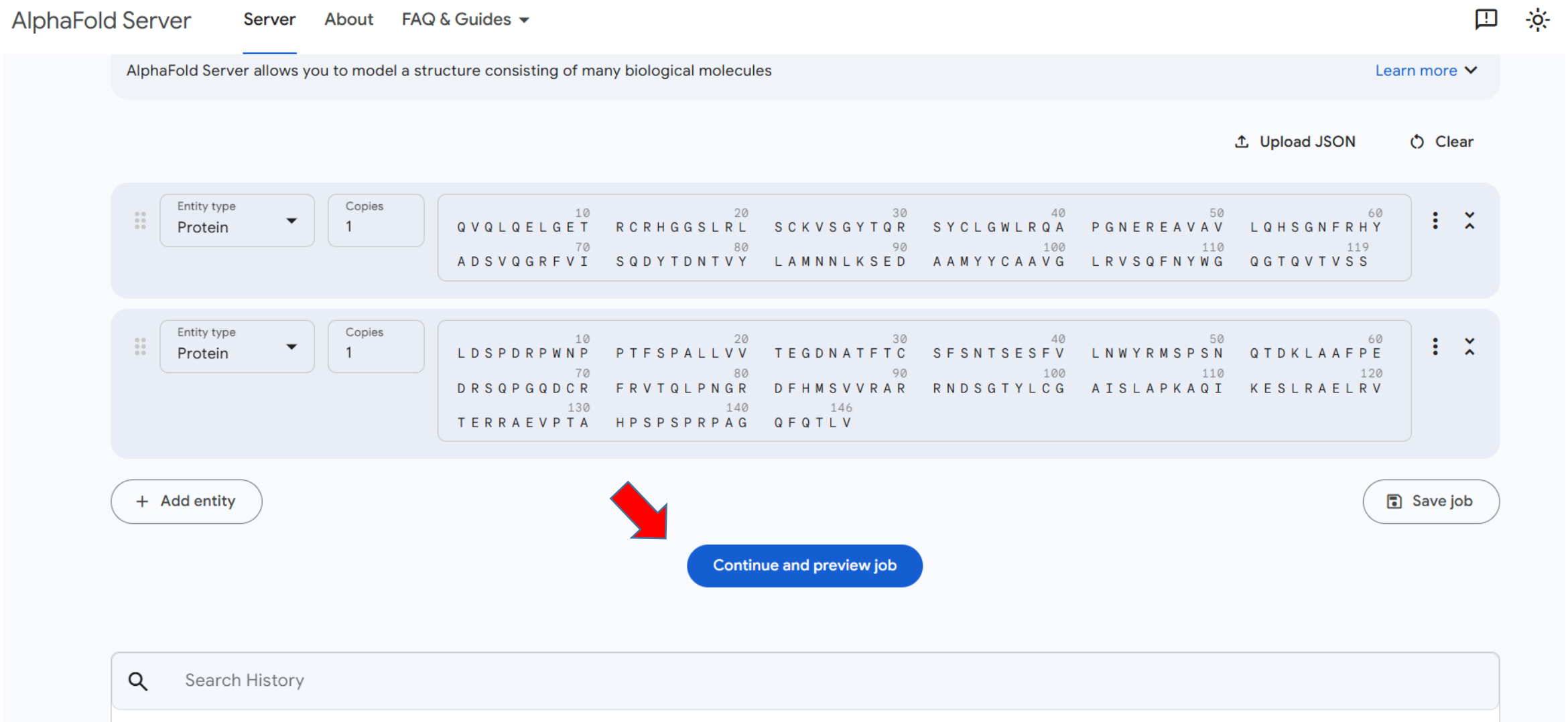


https://www.uniprot.org/



https://alphafoldserver.com/

Nanobody + Extracellular PD1



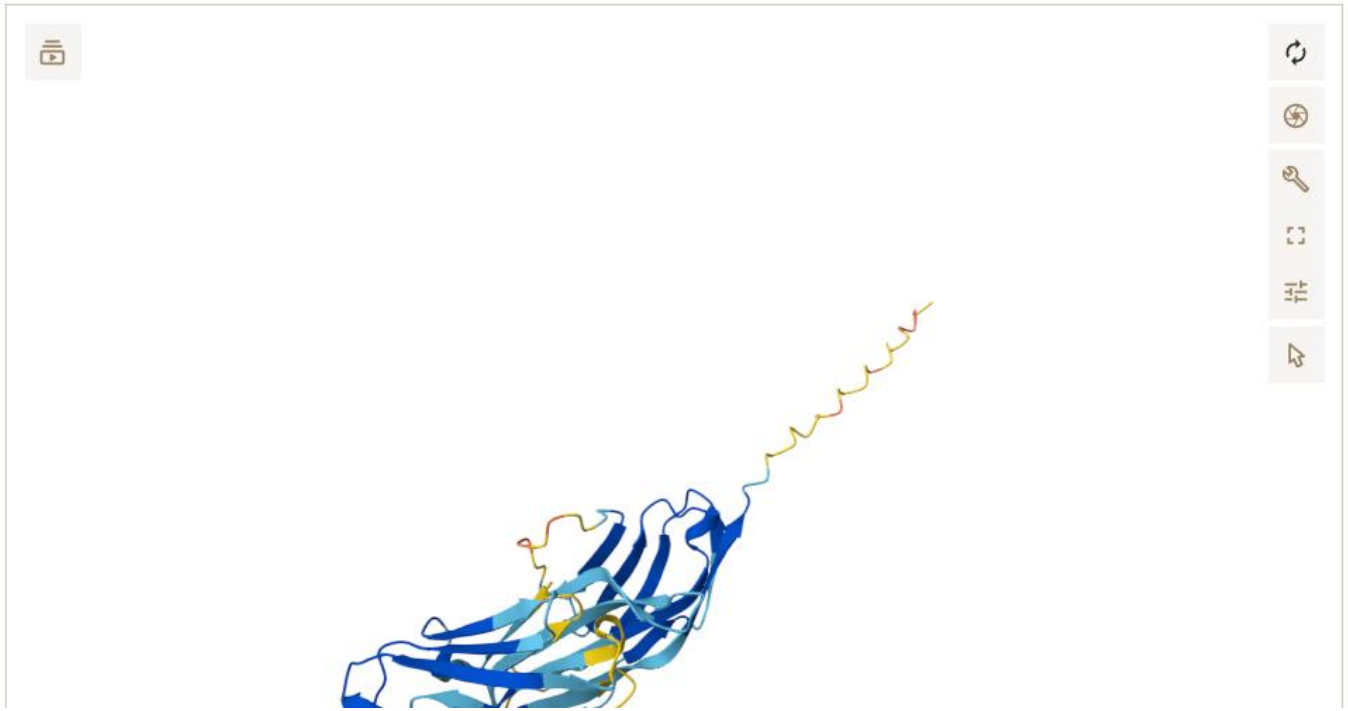
AlphaFold Server Server About FAQ & Guides

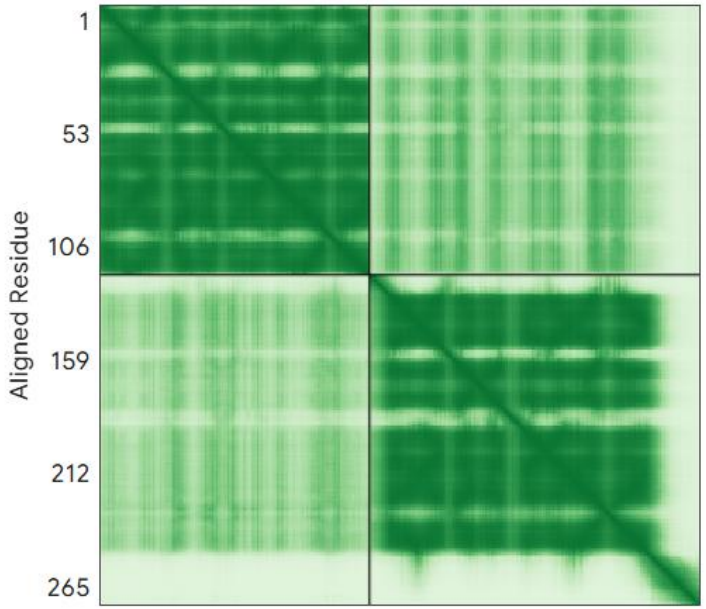
C12_nanobody_interaxtion_PD1_extracellular

Back Download Clone and reuse Feedback on structure

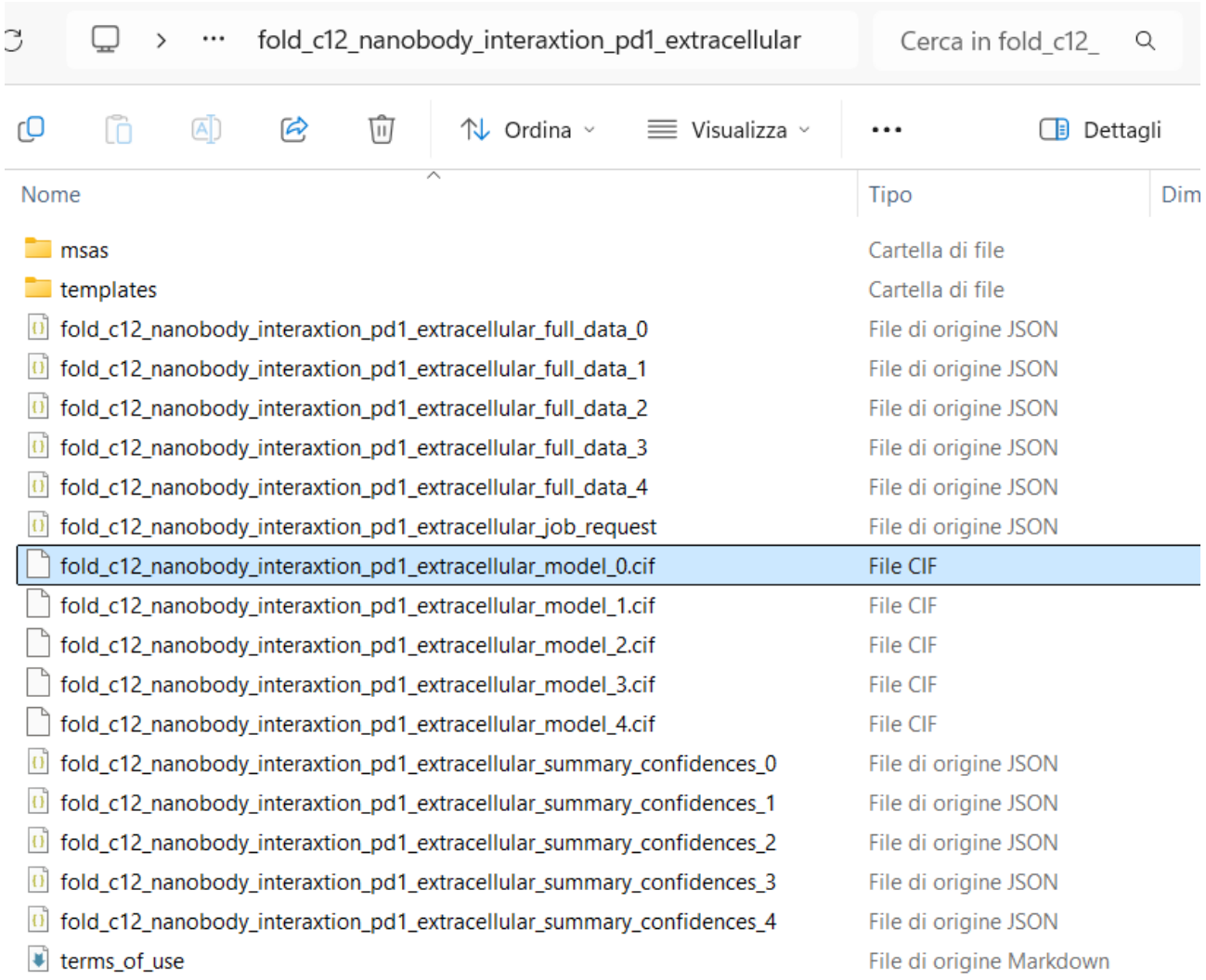
Very high (plDDT > 90) | Confident (90 > plDDT > 70) | Low (70 > plDDT > 50) | Very low (plDDT < 50)

ipTM = 0.23 pTM = 0.52 learn more

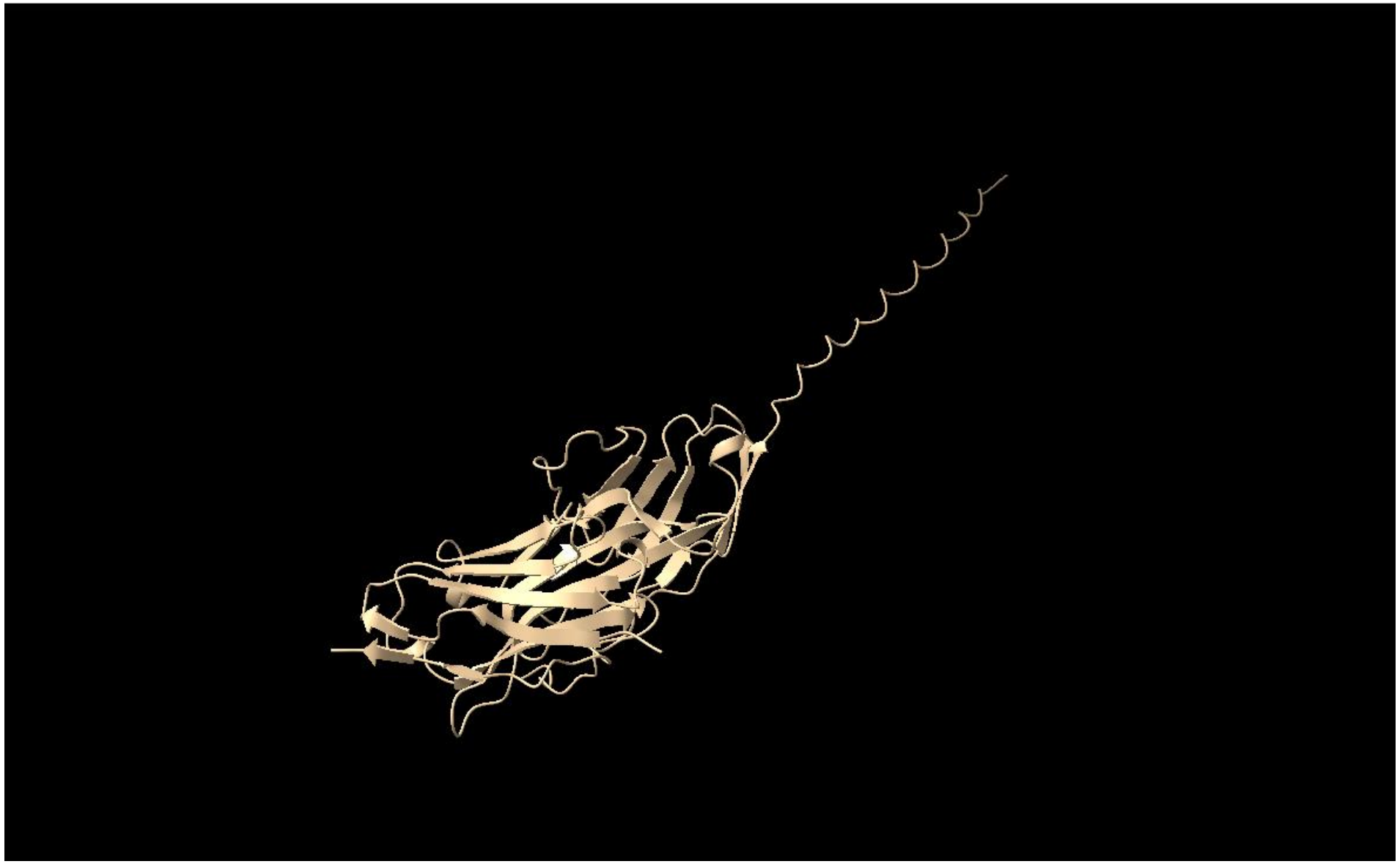




Step 1 - Open this file into ChimeraX

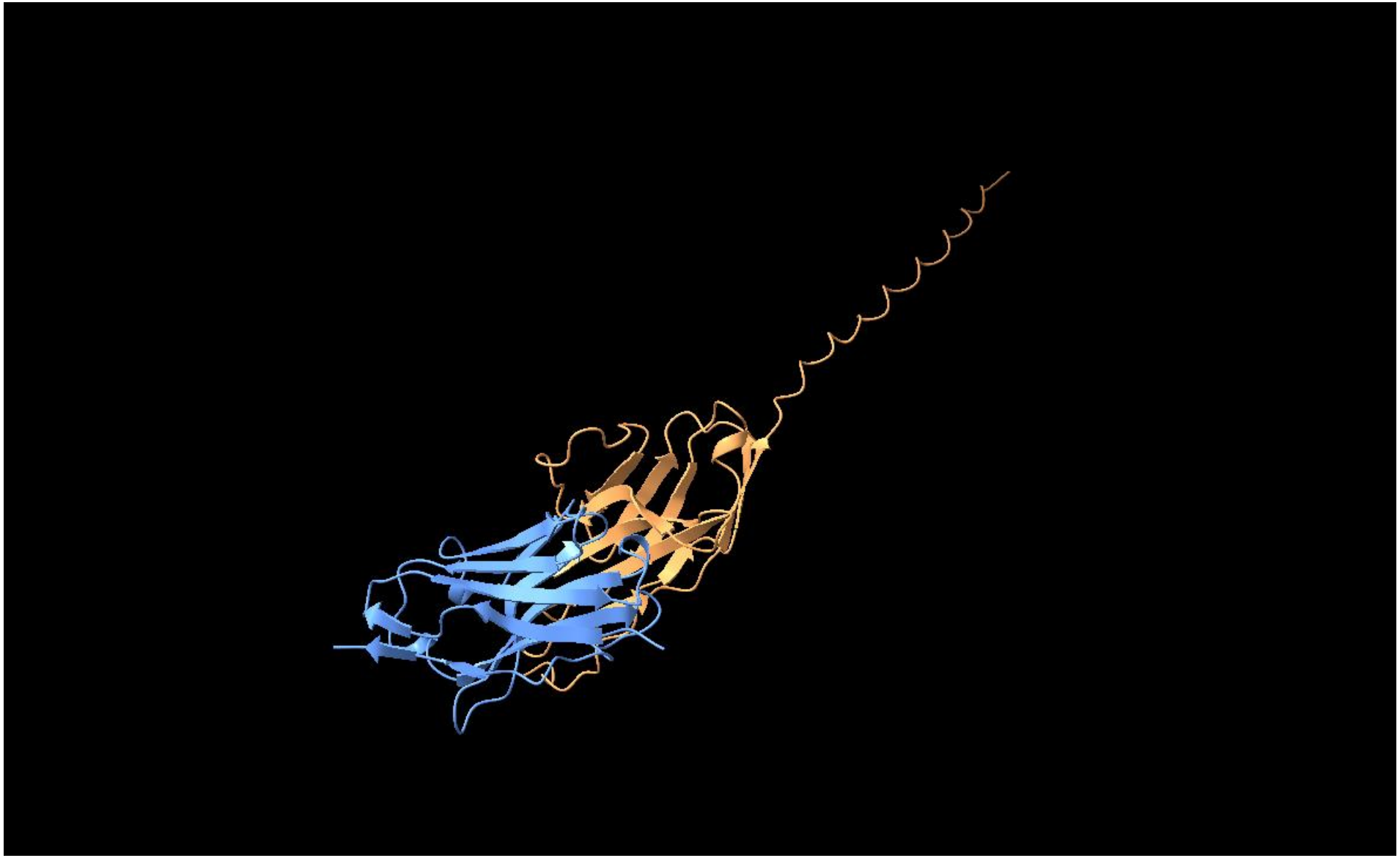


ChimeraX



STEP 2 — Distinguish the two chains

```
# Color Chain A (Nanobody C12) in BLUE
color /A cornflower blue
# Color Chain B (PD-1 extracellular) in ORANGE
color /B sandy brown
```



STEP 3 — Visualize the CDR regions of C12

```
# CDR1 of C12 Nanobody (approx residues 26-35) → CYAN
color /A:26-35 cyan
label /A:26-35 text "CDR1" size 16 color cyan

# CDR2 of C12 Nanobody (approx residues 50-56) → RED
color /A:50-56 red
label /A:50-56 text "CDR2" size 16 color red

# CDR3 of C12 Nanobody (approx residues 95-108) → GREEN
color /A:95-108 green
label /A:95-108 text "CDR3" size 16 color green
```


CDR2
CDR3
CDR1
CDR2
CDR3
CDR3
CDR3
CDR2
CDR1
CDR3
CDR3
CDR3
CDR1
CDR3
CDR3
CDR3
CDR3
CDR1

STEP 4 — Analyze the interface

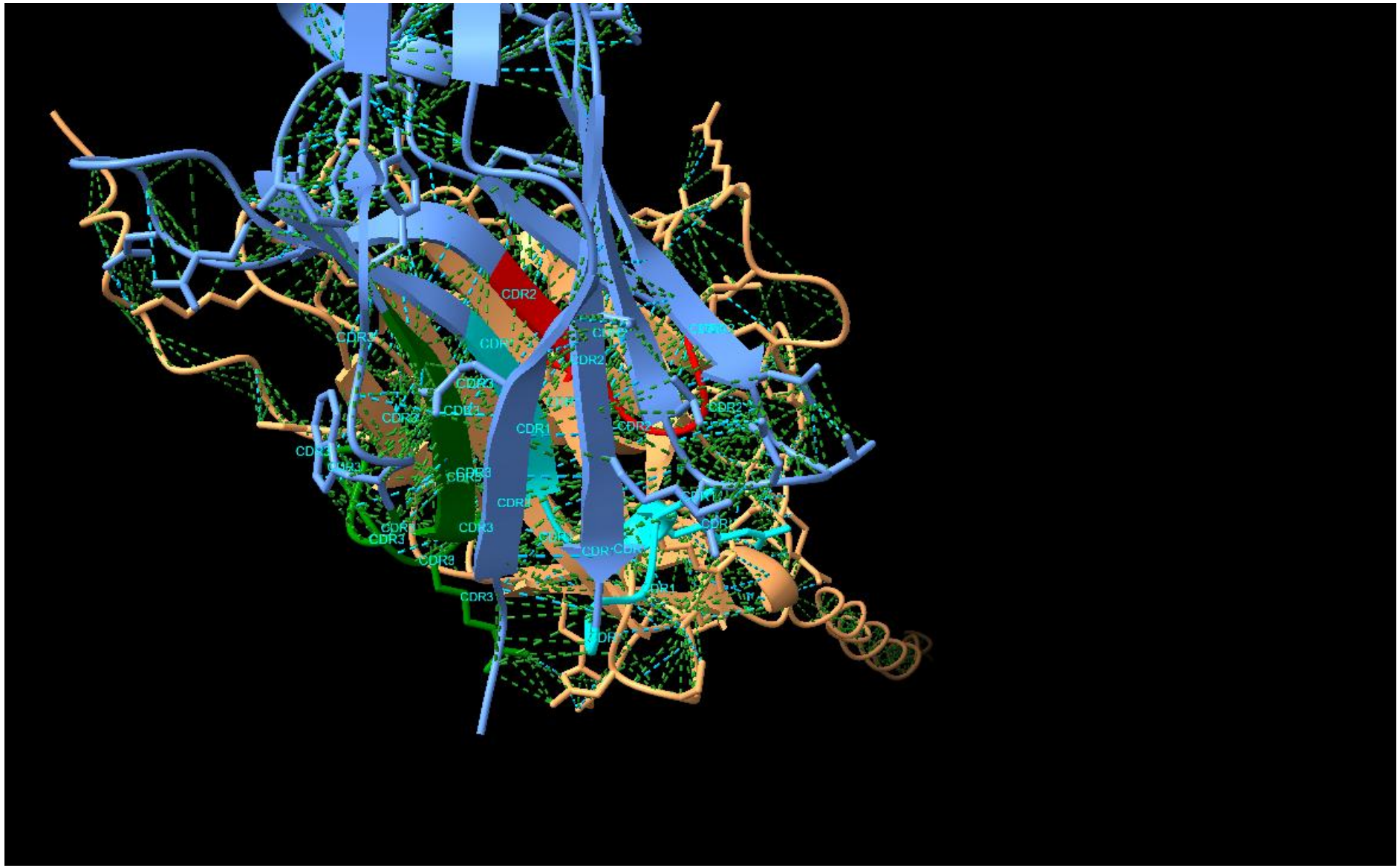
```
# Show hydrogen bonds between the two chains
hbonds /A /B reveal true

# Find interface contacts (within 4 Angstrom)
contacts /A /B distance 4.0

# Show surface of PD-1 to see binding site
surface /B
transparency /B 50

# Focus on the interaction zone
view /A:95-108    # zoom on CDR3 interacting with PD-1
```


CDR2
CDR3
CDR1
CDR2
CDR3
CDR1
CDR2
CDR3
CDR3
CDR3
CDR3
CDR3
CDR3
CDR3
CDR1
CDR2
CDR1

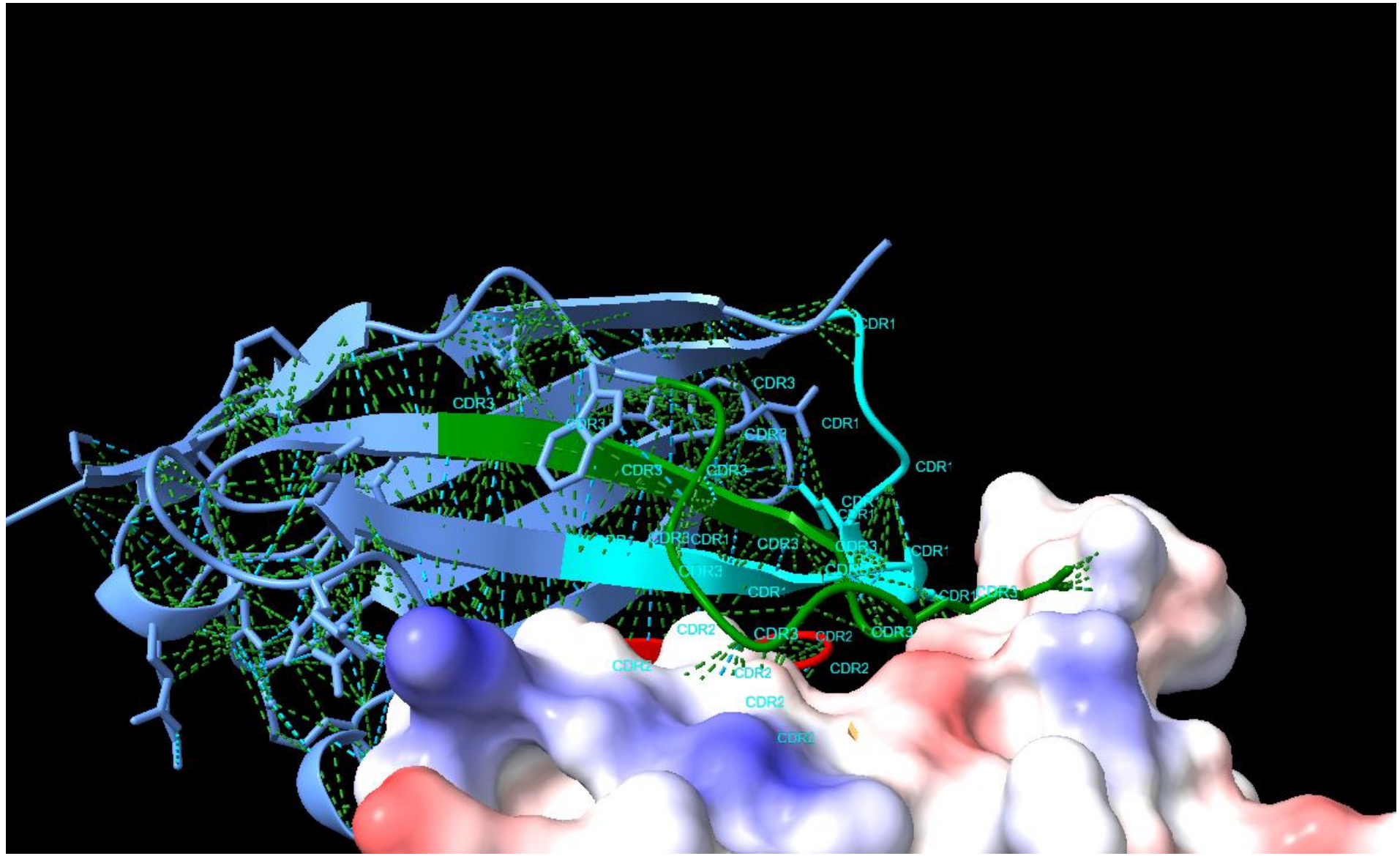

CDR2
CDR3
CDR1
CDR2
CDR3
CDR2
CDR1
CDR2
CDR3
CDR3
CDR1
CDR3
CDR3
CDR3
CDR3

STEP 5 — Display options

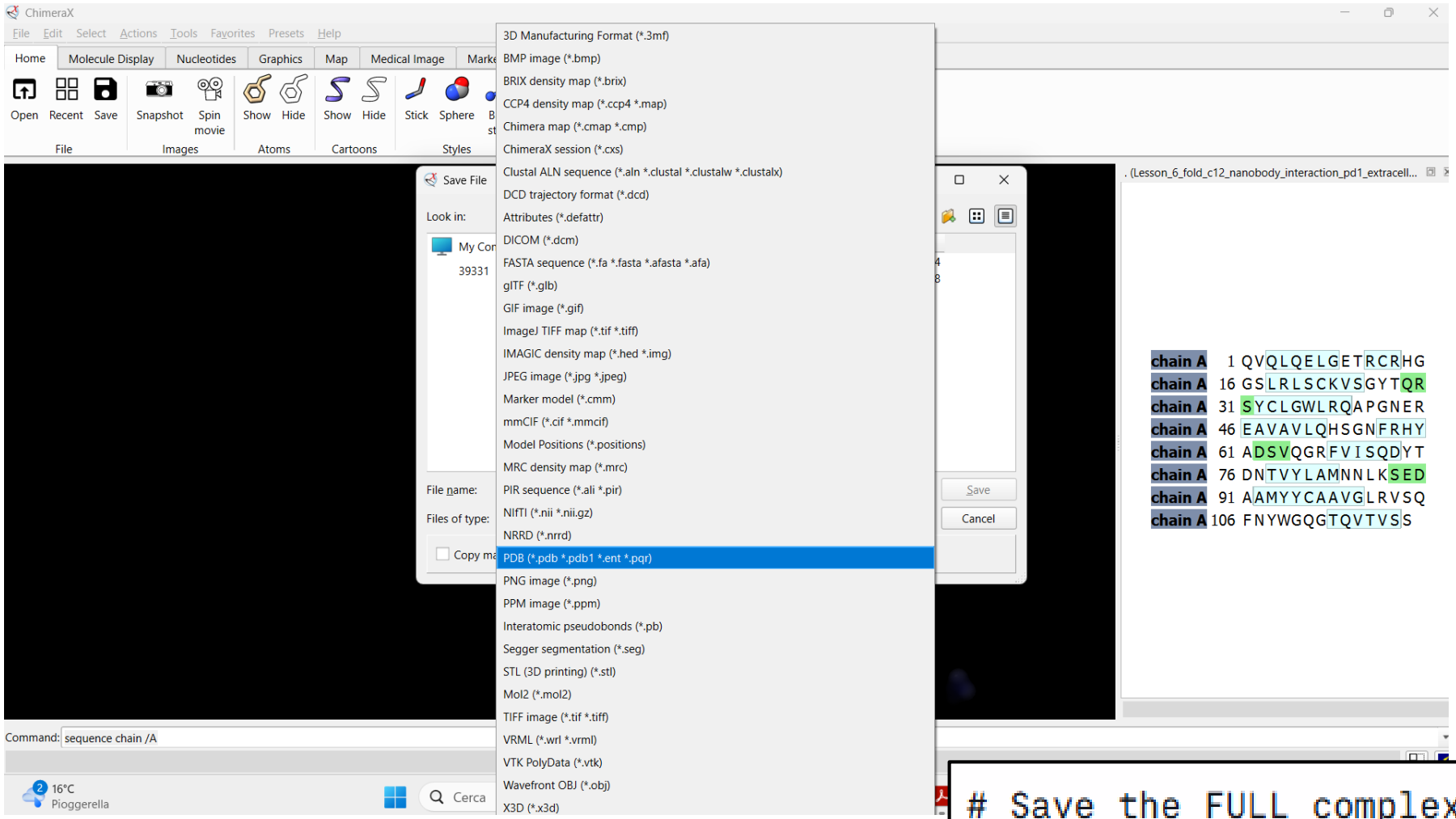
```
# Display sequence viewer (great for teaching)
sequence chain /A
sequence chain /B

# Show electrostatic surface of PD-1
coulombic /B
```

Color	Meaning	Cause
🔵 **Blue**	Positive charge	Lysine (K), Arginine (R), Histidine (H)
🔴 **Red**	Negative charge	Aspartate (D), Glutamate (E)
⚪ **White**	Neutral / hydrophobic	Glycine, Alanine, etc.

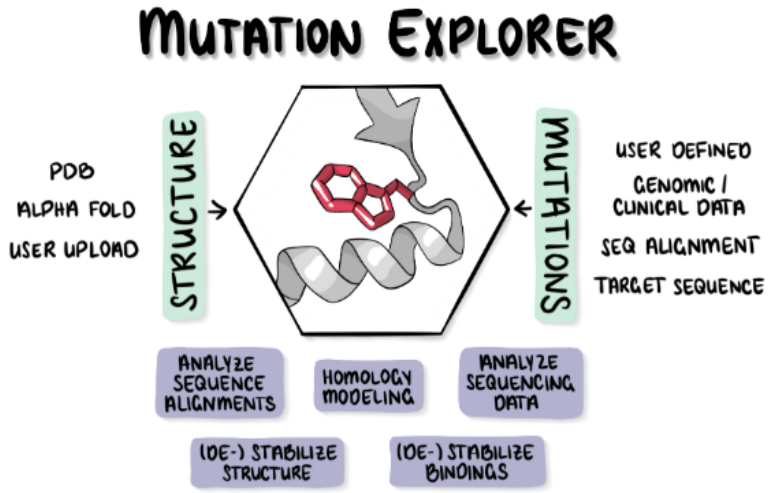

CDR1
CDR3
CDR3
CDR3
CDR1
CDR3
CDR3
CDR1
CDR3
CDR1
CDR3
CDR3
CDR1
CDR2
CDR3
CDR2
CDR3
CDR2
CDR2
CDR2
CDR2
CDR2

STEP 6 — Save as .pdb




```
# Save the FULL complex (both chains A and B)
save C12_PD1_complex.pdb
```

https://mutationexplorer.vda-group.de/mutation_explorer/



The server lets users mutate proteins in various ways and study mutation effects in 3D. You can evolve a protein over many mutation rounds within the viewer under consideration of the mutational effect on the structure. The only required inputs are a structural model and at least one mutation of interest or a target sequence/alignment. Alternatively, the user can upload human SNPs in the VCF format and obtains a visualization of the (de-)stabilizing effect of those mutations on the protein structure.

Go ahead & try it out! You can easily just try out the server and explore or follow one of our tutorials.

If you don't know what input is expected, you can click on the i buttons at each field in the forms to get further information.

Find several tutorials about the usage of the server in our tutorials tab. For more detailed explanation about the methods and the server, incl. interpretation of the results, check out the documentation. Frequently asked questions can be found in the FAQ. If you want to reach out to us, you find the information at the tab "Contacts".

Mutation explorer

Upload structure or model

Choose one of the following options:

Upload a PDB file

C12_PD1_complex.pdb

Load structure from PDB

PDB ID (without '.pdb' ending, example: '1afo')

Load prediction from AlphaFold DB

Uniprot ID

Get notified after job completion

Email (optional, recommended for long minimization)

filter structure

Get notified after job completion

Email (optional, recommended for long minimization)

filter structure

Minimization

i

- ○ long (5-60 min, recommended)
- ● quick (1-3 min)
- ○ none

RaSP Model

i

☐ Calculate RaSP model (~10 min for each chain)

Next: input mutations

load example

Mutation explorer

Define mutations

Select (click on) one or more of the following options:

Manual mutation definition

available PDB chains

info

one letter code

Enter a comma separated list of mutations to be performed together

X:123Y, A:45E

Mutation explorer

Interface Score Definition

Do you want to calculate the interface score?

info

- ○ No calculation
- ● Define manual calculation

Left	Chain	Right
●	A	○
○	B	●

- ○ Complete calculation

submit and launch explorer

Mutation 1

Info

Total energy:	-237.37
Total diff to parent:	1.69
Parent:	mut_0.pdb
Mutations:	A:102A
Original filename:	C12_PD1_complex.pdb
Exploration:	801429

Mutation 2

Info

Total energy:	-229.05
Total diff to parent:	-6.63
Parent:	mut_0.pdb
Mutations:	A:104R
Original filename:	C12_PD1_complex.pdb
Exploration:	801429

Mutation 3

Info

Total energy:	-227.32
Total diff to parent:	-8.36
Parent:	mut_0.pdb
Mutations:	A:108W
Original filename:	C12_PD1_complex.pdb
Exploration:	801429

Mutation	Concept	Expected ΔΔG
`A:102A`	Removing a positive charge	Positive (destabilizing) ❌
`A:104R`	Adding a positive charge	Negative (stabilizing) ✅
`A:108W`	Enhancing aromatic contact	Negative (stabilizing) ✅